An invariance principle for reflected random walk on cell configurations with spatial singularities

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Revised geometric-topology and averaging argument

Abstract

We prove a quenched invariance principle for random walk reflected off of infinity on random cell configurations which may accumulate on, and fail to cover, a closed planar set of zero one-dimensional Hausdorff measure. The adjacency relation is an arbitrary symmetric relation between distinct intersecting cells. Away from an admissible singular-set witness we assume spatial local finiteness and connectedness of the induced cell graph along horizontal and vertical line segments. The probabilistic hypotheses are the scale-covariant mass-transport identity, ergodicity modulo translations and positive dilations, and the two diameter–conductance moments of Gwynne–Miller–Sheffield.

The proof has two features which are useful beyond the present theorem. First, the environment topology is an extension of the Gwynne–Miller–Sheffield whole-cell homeomorphism topology: on the spatially locally finite class the two topologies, and hence their Borel structures, agree exactly. Second, all spatial and temporal dyadic averaging is reduced to one elementary Markov-kernel projection lemma. This avoids any assertion that the environment together with an independent dyadic grid is jointly ergodic and makes the reverse-martingale steps literal conditional-expectation statements.

We construct a finite-specific-energy harmonic coordinate, prove that its corrector is uniformly sublinear, and prove that the canonical reflected path never reaches an end whose cells escape to Euclidean infinity. After the area time change the harmonic coordinate is a local martingale whose predictable bracket is the graph jump bracket; there is no additional continuous bracket at reflection times. A scale-free temporal ergodic theorem then yields a deterministic positive-definite bracket limit, and a martingale functional central limit theorem gives Brownian motion. The same limit holds for deterministic holding times $\mathrm{Area}(H)/\pi(H)$. When the singular set is empty, the hypotheses and topology specialize to those of Gwynne–Miller–Sheffield Theorem 1.16; the face specialization gives their Theorem 1.5.

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1 Environment and main results

We identify $\mathbb{C}$ with $\mathbb{R}^2$. All graph energies below sum over unoriented edges once. Similarities are translations followed by positive dilations; rotations are not included.

1.1 Cell configurations and admissible witnesses

Definition 1.1 (Unmarked cell configuration)**.** *An unmarked cell configuration is a triple $\mathcal{H} = (\mathscr{H}, \sim, c)$ where $\mathscr{H}$ is a countably infinite collection of compact connected subsets of $\mathbb{C}$, called cells, and the following hold.*

(i) Every $H \in \mathscr{H}$ has nonempty interior, and

$$\operatorname{Area}(H \cap K) = 0, \qquad H \neq K.$$

(ii) There exists a closed set $S \subset \mathbb{C}$ such that $\mathcal{H}^1(S) = 0$, every point of $\mathbb{C} \setminus S$ belongs to a cell, and every $z \in \mathbb{C} \setminus S$ has a neighborhood meeting only finitely many cells.

(iii) The relation $\sim$ is symmetric and

$$H \sim K \quad \Longrightarrow \quad H \neq K, \qquad H \cap K \neq \varnothing.$$

For every adjacent pair, $c(H, K) = c(K, H) \in (0, \infty)$.

For $A \subset \mathbb{C}$, $\mathcal{H}(A)$ denotes the induced weighted graph on the cells meeting A.

A closed set S as in Definition 1.1(ii) is called a *geometric witness*. It is *admissible* if, in addition,

$$\mathcal{H}(L) \text{ is connected for every compact horizontal or vertical segment } L \subset \mathbb{C} \setminus S. \tag{LCS}$$

Connectedness always means connectedness by a finite graph path. Intersecting cells need not be adjacent, and the graph need not be planar.

The witness is deliberately not an environment mark. This point is separated from measurability as follows.

Assumption 1.2 (Pathwise witness convention)**.** *The law of $\mathcal{H}$ is carried by a similarity-invariant Borel set Ω_{geom} of environments such that every $\mathcal{H} \in \Omega_{\mathrm{geom}}$ admits at least one admissible witness. No measurable selection $\mathcal{H} \mapsto S(\mathcal{H})$ is assumed or used.*

Thus every argument involving S is deterministic after an environment has been fixed. All random variables, transports, invariant events, correctors, and conditional path laws are functions of the unmarked environment. This formulation avoids any hidden assertion that the existential witness or the simultaneous condition (LCS) is itself a Borel coordinate.

For a cell H write

$$a_H = \operatorname{Area}(H), \qquad d_H = \operatorname{diam}(H), \qquad b_H = \frac{1}{a_H} \int_H z \, dz, \tag{2}$$

and

$$\pi(H) = \sum_{K \sim H} c(H, K), \qquad \pi^*(H) = \sum_{K \sim H} c(H, K)^{-1}, \tag{3}$$

initially as extended nonnegative sums. Since b_H lies in the convex hull of H,

$$|b_H - z| \leq d_H \quad (z \in H), \qquad |b_H - b_K| \leq d_H + d_K \quad (H \sim K). \tag{4}$$

1.2 Topology, mass transport, and moments

Section 2 defines a metric d_{sing} on the unmarked class of Definition 1.1; write $\mathcal{B}_{\mathrm{sing}}$ for its Borel σ-algebra. The restriction of this topology to the spatially locally finite class is exactly the cell-configuration topology of Gwynne–Miller–Sheffield (GMS) [1], not merely a comparable topology; see Proposition 2.5.

If H_z is the unique cell whose interior contains z, a rooted-cell functional is set equal to zero when the root is uncovered or lies on a cell boundary. This convention is similarity covariant, and Lemma 3.4 shows that it is irrelevant at the origin almost surely.

Definition 1.3 (Mass transport modulo scaling)**.** *The law of $\mathcal{H}$ satisfies mass transport modulo scaling if, for every nonnegative $\mathcal{B}_{\mathrm{sing}} \otimes \mathcal{B}(\mathbb{C}^2)$-measurable T satisfying*

$$T(C(\mathcal{H}-u), C(w-u), C(z-u)) = C^{-2} T(\mathcal{H}, w, z), \tag{5}$$

we have

$$\mathbb{E}\left[\int_{\mathbb{C}} T(\mathcal{H}, 0, z)\, dz\right] = \mathbb{E}\left[\int_{\mathbb{C}} T(\mathcal{H}, z, 0)\, dz\right]. \tag{6}$$

We assume the GMS finite-energy moment

$$M := \mathbb{E}\left[\frac{d_{H_0}^2}{a_{H_0}}\big(\pi(H_0) + \pi^*(H_0)\big)\right] < \infty. \tag{FE}$$

Before graph local finiteness is proved, the sums in (FE) are extended sums.

For a weighted graph G define

$$\mathcal{E}_G(f) = \sum_{\{H,K\}\in E(G)} c(H,K)|f(K) - f(H)|^2 \in [0,\infty], \tag{8}$$

and, for finite-energy real- or vector-valued f, g,

$$\mathcal{E}_G(f,g) = \sum_{\{H,K\}\in E(G)} c(H,K)(f(K) - f(H)) \cdot (g(K) - g(H)). \tag{9}$$

The mixed sum is absolutely convergent by Cauchy–Schwarz.

For a bounded closed axis-parallel rectangle Q with nonempty interior, let

$$G_Q = \mathcal{H}(Q), \qquad A_Q = \mathcal{H}(\partial Q),$$

and

$$\mathcal{V}_Q = \{u : V(G_Q) \to \mathbb{R}^2 : u|_{A_Q} = 0,\ \mathcal{E}_{G_Q}(u) < \infty\}. \tag{10}$$

No finite-support condition is imposed.

1.3 Main theorems

Theorem 1.4 (Harmonic coordinate)**.** *Assume Definition 1.1, Assumption 1.2, mass transport (6), and (FE). There exists an unmarked measurable map $\Phi = \Phi_{\mathcal{H}} : \mathscr{H} \to \mathbb{R}^2$, normalized by $\Phi(H_0) = 0$, such that almost surely:*

(a) for every bounded closed axis-parallel rectangle Q,

$$\mathcal{E}_{G_Q}(\Phi) < \infty, \qquad \mathcal{E}_{G_Q}(\Phi, u) = 0 \quad \forall u \in \mathcal{V}_Q; \tag{11}$$

thus Φ is discrete harmonic at every vertex;

(b) its gradient has finite specific energy,

$$\mathbb{E}\left[\frac{1}{2a_{H_0}}\sum_{H\sim H_0} c(H_0,H)|\Phi(H)-\Phi(H_0)|^2\right]<\infty; \tag{12}$$

its gradient commutes with translations and positive dilations;

(c) the corrector is uniformly sublinear,

$$\lim_{R\to\infty}\frac{1}{R}\sup_{H\in\mathcal{H}(B_R(0))}|\Phi(H)-b_H|=0; \tag{13}$$

indeed b_H may be replaced by any choice $z_H\in H$;

(d) the blockwise minimizers of Section 6 converge to Φ in specific energy and, after normalization, in probability on every fixed finite graph neighborhood; a deterministic subsequence converges almost surely in energy on every bounded spatial patch and pointwise at every vertex.

For the process theorem assume additionally that every $\mathcal{B}_{\mathrm{sing}}$-measurable event invariant under translations and positive dilations has probability zero or one. Fix a measurable similarity-covariant representative $z_H\in H$; for example, take the lexicographically first point minimizing distance to b_H.

Theorem 1.5 (Reflected invariance principle)**.** *Under the hypotheses above, conditionally on almost every environment, the following hold from every fixed starting cell.*

(a) There is a canonical reflected ordered path with the area clock X: at a vertex H its transition rate to $K\sim H$ is $c(H,K)/a_H$, so the mean holding time is $a_H/\pi(H)$. It is defined for all positive times, is recurrent at every vertex, and is a homeomorphic time change of an admissible fast reflected walk.

(b) Every graph end ω has a unique spatial image $p_\omega\in S\cup\{\infty\}$ for every admissible witness S, the image is independent of the witness, and the end-to-sphere map is continuous. The spatial vertex path z_{X_t} has a unique càdlàg extension Z, continuous at end-valued times, and the path never attains an end whose spatial image is ∞.

(c) $M_t:=\Phi(X_t)$ has the analogous extension and is a locally square-integrable local martingale. Its predictable bracket is

$$\langle M^i,M^j\rangle_t=\int_0^t\Gamma_{ij}(\mathcal{H},X_s)\,ds,\qquad \Gamma(\mathcal{H},H)=\frac{1}{a_H}\sum_{K\sim H}c(H,K)\,\Delta\Phi\,\Delta\Phi^{\mathsf{T}}, \tag{14}$$

where $\Delta\Phi=\Phi(K)-\Phi(H)$ and the integrand is zero at nonvertex times.

(d) With

$$\Sigma=\mathbb{E}[\Gamma(\mathcal{H},H_0)], \tag{15}$$

Σ is finite and positive definite. If $\widetilde{Z}$ denotes the path obtained by linearly interpolating each ordinary graph jump over the preceding holding interval and by using the end-time limits at reflection times, then

$$\mathcal{L}\big((\varepsilon\widetilde{Z}_{t/\varepsilon^2})_{t\ge0}\mid\mathcal{H}\big)\Longrightarrow\mathcal{L}((B_t)_{t\ge0}) \tag{16}$$

in $C([0,\infty),\mathbb{R}^2)$ with local uniform topology, where B is Brownian motion with $\mathbb{E}[B_tB_t^{\mathsf{T}}]=t\Sigma$.

(e) The same limit, with the same Σ, holds for the same ordered reflected path when every visit to H is assigned the deterministic holding time $a_H/\pi(H)$, and for any other measurable choice of representatives inside the cells.

No uniformity over starting cells depending on ε is asserted.

1.4 External reflected-walk input

Only the reflected-walk construction and standard Dirichlet-form facts are imported. To avoid ambiguity about hypotheses, we record exactly what is used.

Proposition 1.6 (Reflected-walk input)**.** *Let G be a connected countable conductance network. If the ordinary conductance walk is recurrent, use that walk with any positive finite rate function. If it is transient, use the Gwynne–Sung reflected walk [2] with an admissible rate $w \geq w_*$. In either case there is a recurrent continuous-time process on $V(G) \cup \{\infty\}$ with the following properties.*

(i) At vertex times it has ordinary conductance transition probabilities and positive holding intervals. It has the vertex-valued strong Markov property at the departure, hitting, and return times used below, and it returns arbitrarily late to every fixed vertex.

(ii) For every nonempty finite $A \subset V(G)$ and boundary data $\varphi : A \to \mathbb{R}$, the first-hit expectation is the unique finite-energy extension minimizing graph energy among functions agreeing with φ on A.

(iii) If G is locally finite, the path has the fixed-finite-set regularity and finite-separator property of Gwynne–Sung: before hitting a finite vertex set it cannot pass between distinct components of its complement. Every nonvertex time has a unique graph-end label, with the local end-neighborhood behavior of their Proposition 5.3.

(iv) For a fast rate w with $m = \pi/w$ summable, the symmetric form of the process on $L^2(m)$ is the active reflected form

$$\mathcal{F}^{\mathrm{ref}} \cap L^2(m), \qquad \mathcal{F}^{\mathrm{ref}} = \{f : \mathcal{E}_G(f) < \infty\},$$

with quadratic form equal to graph energy.

For the transient case, these are the existence/uniqueness theorem, the finite-target hitting statement, Proposition 3.15, the fixed-vertex path lemmas, and Proposition 5.3 of Gwynne–Sung. In the recurrent case the corresponding statements reduce to the ordinary recurrent conductance walk. Standard regular symmetric Dirichlet-form theory [3, 4] is used only after Proposition 1.6(iv) has identified the abstract form.

2 The geometric environment topology

Let $\mathcal{C}_{\mathrm{sing}}$ be the unmarked class of Definition 1.1; no line-connectivity hypothesis is included in the definition of this space. For $\mathcal{H} \in \mathcal{C}_{\mathrm{sing}}$ define the intrinsic accumulation set

$$S_{\mathrm{acc}}(\mathcal{H}) = \{z \in \mathbb{C} : \#\mathcal{H}(B_r(z)) = \infty \text{ for every } r > 0\}. \tag{17}$$

It is closed and is contained in every geometric witness, hence $\mathcal{H}^1(S_{\mathrm{acc}}) = 0$. Every compact subset of its complement meets only finitely many cells.

2.1 Finite restrictions

A finite restriction is a finite collection of whole cells with its induced adjacency relation and conductances. The empty restriction is allowed. If F, F' are finite restrictions, define

$$\delta(F, F') = 1 \wedge \inf_f \left\{ \|f - \mathrm{id}\|_\infty + \max_{\{H,K\} \in E(F)} |c_F(H, K) - c_{F'}(f(H), f(K))| \right\}, \tag{18}$$

where the infimum is over plane homeomorphisms mapping the cells of F bijectively to the cells of F' and preserving adjacency in both directions. The maximum over an empty edge set is zero, and the infimum over an empty set is $+\infty$. Add a symbol $\dagger$ with $\delta(\dagger,\dagger)=0$ and $\delta(\dagger,F)=1$ for ordinary restrictions.

Fix an enumeration $(q^{(j)})_{j\ge 1}$ of all nonempty finite tuples of points of $\mathbb{Q}^2$. If $q=(q_1,\dots,q_k)$, set

$$U_{q,r}=\bigcup_{i=1}^{k} B_r(q_i), \qquad r>0.$$

For an open set U and $N\ge 1$ define

$$P_N(\mathcal{H},U)=\begin{cases}\mathcal{H}(U), & \#\mathcal{H}(U)\le N,\\ \dagger, & \#\mathcal{H}(U)>N.\end{cases}$$

Definition 2.1 (Geometric topology)**.** *Set*

$$d_{\mathrm{sing}}(\mathcal{H},\mathcal{H}')=\sum_{j,N\ge 1} 2^{-j-N}\int_0^\infty e^{-r}\,\delta\big(P_N(\mathcal{H},U_{q^{(j)},r}),P_N(\mathcal{H}',U_{q^{(j)},r})\big)\,dr. \tag{19}$$

Write τ_{sing} for the induced topology and $\mathcal{B}_{\mathrm{sing}}$ for its Borel σ-algebra.

The rational points specify test windows; they are not cell labels. Finite unions are needed because two cells visible in separated regular neighborhoods can be adjacent through a point of S_{acc}.

Lemma 2.2. *The functions δ and d_{sing} are metrics on their respective spaces. The topology of d_{sing} is unchanged if the enumeration of rational tuples is changed or if the weights 2^{-j-N} are replaced by arbitrary strictly positive summable weights.*

Proof. Symmetry and the triangle inequality for δ follow by inversion and composition of matching homeomorphisms; truncation at one preserves the triangle inequality. If $\delta(F,F')=0$, choose matchings f_n with $\|f_n-\mathrm{id}\|_\infty\to 0$ and vanishing conductance error. An interior point of any $H\in F$ belongs, for all large n, to the interior of the same cell of F' under f_n, because distinct cells have disjoint interiors. Hausdorff convergence then identifies that cell with H. Reversing the roles of F,F' gives equality of the restrictions, including adjacency and conductances.

For fixed $\mathcal{H},q,N$, the map $r\mapsto P_N(\mathcal{H},U_{q,r})$ changes only when a new cell enters, until the count exceeds N, after which it is constantly $\dagger$. Thus the integrand in (19) is measurable. Positivity, symmetry, and the triangle inequality follow termwise.

For separation, fix finitely many cells of $\mathcal{H}$ and choose rational points in their interiors outside $S_{\mathrm{acc}}(\mathcal{H})$. For some interval (a,b) the corresponding union of closed radius-b disks avoids S_{acc}, hence meets only finitely many cells, and all selected cells occur in the restriction for every $r\in(a,b)$. If $d_{\mathrm{sing}}(\mathcal{H},\mathcal{H}')=0$, then for almost every such r the corresponding finite restrictions have δ-distance zero. Hence the selected cells, their mutual adjacencies, and conductances occur in $\mathcal{H}'$. Reverse the roles of $\mathcal{H},\mathcal{H}'$. Since the finite list was arbitrary, the full triples agree. The assertion about the weights follows by truncating the summable tail. □

2.2 Local matching and stability

Lemma 2.3 (Local whole-cell matching)**.** *Suppose $\mathcal{H}_n\to\mathcal{H}$ in d_{sing}. Let $K\subset\mathbb{C}\setminus S_{\mathrm{acc}}(\mathcal{H})$ be compact and let F be a finite set of cells of $\mathcal{H}$. Then there are bounded open sets $V_n\supset K$ and plane homeomorphisms f_n such that, for all large n,*

$$F\subset\mathcal{H}(V_n), \qquad f_n:\mathcal{H}(V_n)\longrightarrow\mathcal{H}_n(V_n) \tag{20}$$

is an adjacency-preserving bijection and

$$\|f_n - \mathrm{id}\|_\infty + \max_{\{H,H'\}\in E(\mathcal{H}(V_n))} |c(H,H') - c_n(f_n(H), f_n(H'))| \longrightarrow 0.$$

The limiting restrictions are contained in one fixed finite restriction. In particular every cell of $\mathcal{H}_n$ meeting K is among the matched cells.

Proof. Choose a rational regular interior point in each cell of F and finitely many rational regular centers covering K. Since $\mathbb{C} \setminus S_{\mathrm{acc}}$ is open, there are $0 < a < b$ such that

$$K \subset U_{q,a}, \qquad \overline{U_{q,b}} \cap S_{\mathrm{acc}}(\mathcal{H}) = \varnothing, \qquad F \subset \mathcal{H}(U_{q,a}).$$

The restriction $\mathcal{H}(U_{q,b})$ is finite. Choose N at least its cardinality. Convergence of the corresponding term in (19) gives radii $r_n \in (a,b)$ at which the δ-error tends to zero. For large n neither restriction is †. Choose almost-minimizing homeomorphisms and set $V_n = U_{q,r_n}$. □

Lemma 2.4 (Stability of restrictions)**.** *Fix $\mathcal{H}$ and a finite tuple $p = (p_1, \dots, p_k) \in \mathbb{C}^k$. Outside a Lebesgue-null set of $r > 0$, if $\mathcal{H}_n \to \mathcal{H}$, $p_n \to p$, and $r_n \to r$, then for every N,*

$$\delta\big(P_N(\mathcal{H}_n, U_{p_n,r_n}), P_N(\mathcal{H}, U_{p,r})\big) \longrightarrow 0. \tag{21}$$

Proof. Exclude the radii in

$$\bigcup_i \{|z - p_i| : z \in S_{\mathrm{acc}}(\mathcal{H})\} \;\cup\; \{\min_i \mathrm{dist}(p_i, H) : H \in \mathscr{H}\}.$$

The first set is null because distance is 1-Lipschitz and $\mathcal{H}^1(S_{\mathrm{acc}}) = 0$; the second is countable. Put $U = U_{p,r}$. If $\#\mathcal{H}(U) > N$, choose $N+1$ cells meeting U and use Lemma 2.3; their images still meet U_{p_n,r_n} for large n, so both capped restrictions are †.

If $\mathcal{H}(U)$ is finite, the excluded-radius condition gives a compact neighborhood K of $\overline{U}$ disjoint from S_{acc}. Lemma 2.3 matches every cell which can meet the perturbed windows. There are only finitely many limiting cells involved, and for each the strict inequality $\min_i \mathrm{dist}(p_i, H) < r$ or $> r$ is stable under the small homeomorphism and the perturbation $(p_n, r_n) \to (p, r)$. Restrict the matching to the cells which meet the window. □

2.3 Exact agreement with the GMS topology

Let $\mathcal{C}_{\mathrm{GMS}} \subset \mathcal{C}_{\mathrm{sing}}$ be the configurations which are spatially locally finite everywhere. Their cells cover $\mathbb{C}$: locally, a finite union of closed cells contains the dense complement of any geometric witness, hence the whole neighborhood. Therefore this is exactly the unmarked full-plane class of GMS Definition 1.15.

Let $d^{\mathrm{CC}}_{\mathrm{GMS}}$ denote the metric in GMS equation (1.15), i.e.

$$d^{\mathrm{CC}}_{\mathrm{GMS}}(\mathcal{H}, \mathcal{H}') = \int_0^\infty e^{-r} \wedge \inf_f \left\{ \|f - \mathrm{id}\|_\infty + \max_{\{H,K\}\in E(\mathcal{H}(B_r(0)))} |c(H,K) - c'(f(H), f(K))| \right\} dr, \tag{22}$$

where f bijectively matches all whole cells meeting $B_r(0)$ and preserves adjacency in both directions.

Proposition 2.5 (Faithfulness to GMS)**.** *The subspace topology satisfies*

$$\tau_{\mathrm{sing}}|_{\mathcal{C}_{\mathrm{GMS}}} = \tau_{\mathrm{GMS}}. \tag{23}$$

Moreover $\mathcal{C}_{\mathrm{GMS}}$ is a similarity-invariant Borel subset of $\mathcal{C}_{\mathrm{sing}}$. Consequently its trace Borel σ-algebra is exactly the Borel structure used by GMS.

Proof. Suppose first that $d_{\mathrm{sing}}(\mathcal{H}_n, \mathcal{H}) \to 0$ with all environments in $\mathcal{C}_{\mathrm{GMS}}$. By Lemma 2.4, for almost every $r > 0$ the whole-cell restrictions on $B_r(0)$ admit matchings with error tending to zero. These matchings are admissible in (22); dominated convergence gives $d^{\mathrm{CC}}_{\mathrm{GMS}}(\mathcal{H}_n, \mathcal{H}) \to 0$.

Conversely suppose $d^{\mathrm{CC}}_{\mathrm{GMS}}(\mathcal{H}_n, \mathcal{H}) \to 0$. Fix a rational tuple q, a cutoff N, and a radius r which is not a cell-entry radius $\min_i \operatorname{dist}(q_i, H)$ of the limiting configuration. Choose R so large that a compact neighborhood of $\overline{U_{q,r}}$ lies in $B_R(0)$. Since the integral in (22) tends to zero, for each large n there is a radius $\rho_n \in (R, R+1)$ and a whole-cell matching on $B_{\rho_n}(0)$ whose error tends to zero. Spatial local finiteness of $\mathcal{H}$ leaves only finitely many cells capable of changing membership in $U_{q,r}$, and the non-entry-radius condition gives a positive strict-comparison margin for this finite set. For large n the matching therefore restricts to an exact matching of the cells meeting $U_{q,r}$, with the same vanishing error. If the limiting restriction contains more than N cells, the same is eventually true on the approximating side. Hence the integrand in each fixed (q, N) term of (19) tends to zero for almost every r. Dominated convergence in r, followed by truncation of the summable (j, N) tail, yields $d_{\mathrm{sing}}(\mathcal{H}_n, \mathcal{H}) \to 0$. Since both spaces are metrizable, the two sequential implications prove (23).

For an open U, the map $\mathcal{H} \mapsto \#\mathcal{H}(U)$ is lower semicontinuous by Lemma 2.3. Thus

$$\mathcal{C}_{\mathrm{GMS}} = \bigcap_{m \geq 1} \bigcup_{N \geq 1} \{\mathcal{H} : \#\mathcal{H}(B_m(0)) \leq N\}$$

is Borel. Similarities preserve spatial local finiteness. □

The cutoff P_N therefore does not weaken the topology on the original GMS class. It only permits cell counts to diverge in windows approaching an accumulation point.

2.4 Similarity action and Borel coordinates

The following elementary extension lemma fills the only ambient-homeomorphism point needed for continuity of dilations.

Lemma 2.6 (Compactly supported affine correction)**.** *Let $K \subset \mathbb{C}$ be compact and let $A_n(z) = \alpha_n z + \beta_n$ with $\alpha_n > 0$, $\alpha_n \to 1$, and $\beta_n \to 0$. There exist plane homeomorphisms g_n such that $g_n = A_n$ on a neighborhood of K and*

$$\|g_n - \mathrm{id}\|_\infty \longrightarrow 0.$$

Proof. Choose R with $K \subset B_R(0)$ and a smooth cutoff χ equal to one on $B_R(0)$ and zero outside $B_{2R}(0)$. Put

$$g_n(z) = z + \chi(z)\big((\alpha_n - 1)z + \beta_n\big).$$

The perturbation has sup norm tending to zero. Its Lipschitz constant also tends to zero, so for large n it is < 1. Hence $g_n = \mathrm{id} + h_n$ is globally bi-Lipschitz: $|g_n(x) - g_n(y)| \geq (1 - \operatorname{Lip} h_n)|x - y|$, and surjectivity follows from the contraction mapping theorem applied to $x = y - h_n(x)$. Thus g_n is a plane homeomorphism with the stated properties. □

Proposition 2.7 (Continuity of similarities)**.** *The map*

$$((C, u), \mathcal{H}) \longmapsto C(\mathcal{H} - u), \qquad (C, u) \in (0, \infty) \times \mathbb{C},$$

is jointly continuous for τ_{sing}.

Proof. Let $C_n \to C > 0$, $u_n \to u$, and $\mathcal{H}_n \to \mathcal{H}$. A fixed test window $U_{q,r}$ for $C_n(\mathcal{H}_n - u_n)$ pulls back to disks centered at $u_n + q_i/C_n$ with radius r/C_n. For almost every r, Lemma 2.4 gives a small whole-cell matching between the corresponding finite preimage restrictions whenever the limiting count is at most N; if the limiting count exceeds N, the approximating counts eventually do as well.

Conjugate the preimage matching by the fixed similarity $z \mapsto C(z - u)$. On the finitely many matched cells, the remaining discrepancy is the affine map tending to the identity which converts (C, u) to (C_n, u_n). Lemma 2.6 realizes this discrepancy by a compactly supported ambient homeomorphism with uniformly vanishing displacement, equal to the required affine correction on a neighborhood of all matched cells. Thus the transformed finite restrictions have δ-distance tending to zero for almost every r. Dominated convergence and the summable tail in (19) finish the proof. □

Fix an enumeration (q_j) of $\mathbb{Q}^2$. Assign each cell to the first rational point in its interior and put an empty symbol at unused indices. Record the compact cell, adjacency indicators, and conductances.

Proposition 2.8 (Auxiliary Borel coding)**.** *This produces an injective Borel map*

$$J : (\mathcal{C}_{\text{sing}}, \mathcal{B}_{\text{sing}}) \longrightarrow (\mathcal{K}(\mathbb{C}) \sqcup \{\varnothing\})^{\mathbb{N}} \times \{0,1\}^{\mathbb{N}\times\mathbb{N}} \times [0,\infty)^{\mathbb{N}\times\mathbb{N}}.$$

The coding is only a measurable coordinate system; its product topology is not the environment topology.

Proof. For fixed rational q, the set on which q lies in a cell interior is open by Lemma 2.3, and on this set the corresponding whole cell varies continuously in Hausdorff distance. Interior membership of a point in a compact set is Borel, so the event that q_j is the first rational interior point of its cell is Borel. Once finitely many cells are specified by such interior points, Lemma 2.3 shows that adjacency is locally constant and conductance varies continuously under the local matching. Hence every coordinate of J is Borel. Every cell has a rational interior point outside S_{acc}, so every cell is recorded exactly once. □

Corollary 2.9 (Compatibility on the GMS class)**.** *For a law supported on $\mathcal{C}_{\text{GMS}}$, the mass-transport and ergodicity hypotheses in this paper are equivalent to the corresponding GMS hypotheses. In particular, any GMS law satisfying one of their equivalent formulations of translation invariance modulo scaling satisfies (6) here.*

Proof. Proposition 2.5 identifies the subspace Borel structures. Restriction of a transport from $\mathcal{C}_{\text{sing}}$ to $\mathcal{C}_{\text{GMS}}$ is immediate. Conversely, extend a GMS Borel transport by zero off the invariant Borel subspace $\mathcal{C}_{\text{GMS}}$; covariance is preserved. The same restriction/extension argument applies to invariant events. The equivalence with the other GMS formulations is invoked only on this original class. □

3 Geometric consequences of the singular-set hypothesis

Fix an environment in Ω_{geom} and one admissible witness S. The conclusions below do not depend on choosing S measurably.

Lemma 3.1 (Null boundaries)**.** *Distinct cell interiors are disjoint; every cell boundary has zero area; and the uncovered set has zero one-dimensional Hausdorff measure. In particular the cells cover Lebesgue-a.e. point of $\mathbb{C}$.*

Proof. If two interiors intersect, their intersection contains a nonempty open set and hence has positive area, contradicting Definition 1.1(i). The uncovered set is contained in S.

Fix H and $z \in \partial H \setminus S$. A neighborhood of z meets only finitely many cells. Since z is a boundary point, there are points of $\mathbb{C} \setminus H$ arbitrarily close to z; because S has empty interior they may be chosen outside S. A subsequence belongs to one fixed cell $K \neq H$, and closedness gives $z \in K$. Hence

$$\partial H \subset S \cup \bigcup_{K \neq H} (H \cap K),$$

whose area is zero. Finally $\mathcal{H}^1(S) = 0$ implies planar Lebesgue measure zero. □

Lemma 3.2 (Compact sets and good line offsets)**.** *Every compact $K \subset \mathbb{C} \setminus S$ meets finitely many cells and is covered by them. The coordinate projections*

$$N_h = p_y(S), \qquad N_v = p_x(S)$$

are Borel Lebesgue-null subsets of $\mathbb{R}$.

Proof. Local finiteness and compactness give the first assertion. A coordinate projection is 1-Lipschitz, so $\mathcal{H}^1(p_i(S)) \leq \mathcal{H}^1(S) = 0$. Since S is σ-compact, its projections are σ-compact and therefore Borel. □

Proposition 3.3 (Generic lines and global connectedness)**.** *For every $y \notin N_h$ and every real $a < b$, the graph $\mathcal{H}([a, b] \times \{y\})$ is finite and connected; the analogous statement holds for vertical segments at $x \notin N_v$. The full cell graph is connected.*

If the four side offsets of a bounded nondegenerate axis-parallel rectangle Q avoid N_h, N_v, then A_Q is finite, nonempty, and connected, and G_Q is connected.

Proof. The indicated segments avoid S, so finiteness follows from Lemma 3.2 and connectedness from (LCS). To connect arbitrary cells H, K, choose a good horizontal height through H° and a good vertical coordinate through K°. Their intersection lies outside S and therefore in a cell J. The two line segments furnish finite graph paths $H \rightsquigarrow J \rightsquigarrow K$.

For a good rectangle, each side has a finite connected induced graph, and consecutive sides share a cell at their covered common corner. Hence A_Q is finite and connected. A component of G_Q disjoint from A_Q consists of cells contained in Q°. Every neighbor of such a cell still meets Q, so the component has no edge to the complement in the global graph. Global connectedness would force all cells to lie in Q, contradicting almost-everywhere coverage outside the bounded rectangle. □

Lemma 3.4 (The Lebesgue root is regular)**.** *Under mass transport, the origin lies in the interior of a unique cell almost surely.*

Proof. Let $Z(\mathcal{H})$ be any jointly measurable similarity-covariant planar set of zero area. Apply (6) to

$$T(\mathcal{H}, u, v) = \mathbf{1}_{\{u \in Z(\mathcal{H}),\ u \neq v\}} |u - v|^{-2}.$$

The incoming integral at the origin is zero because Z has zero area, whereas the outgoing integral is infinite on $\{0 \in Z\}$. Hence $\mathbb{P}[0 \in Z] = 0$. Apply this to the intrinsic uncovered set and to the union of all cell boundaries, whose membership relations are Borel by Proposition 2.8 and Lemma 3.1. Disjoint interiors give uniqueness. □

Lemma 3.5 (Graph local finiteness)**.** *Under* (FE)*, every cell has finitely many neighbors almost surely. Consequently*

$$0 < \pi(H), \pi^*(H) < \infty$$

for every cell.

Proof. If H has n distinct neighbors with conductances $c_1, \dots, c_n$, then

$$\pi(H) + \pi^*(H) \geq \sum_{k=1}^{n} (c_k + c_k^{-1}) \geq 2n.$$

Thus an infinite-degree root makes the integrand in (FE) infinite, and Lemma 3.4 gives $\mathbb{P}[\deg(H_0) = \infty] = 0$.

Let $D(\mathcal{H})$ be the union of interiors of infinite-degree cells. It is a measurable similarity-covariant open set. The same $|u - v|^{-2}$ transport as in Lemma 3.4 shows that $D(\mathcal{H})$ has zero area almost surely. If an infinite-degree cell existed, $D(\mathcal{H})$ would contain its nonempty interior, a contradiction. Positive degree follows from Proposition 3.3 and the infinitude of the cell collection. □

Proposition 3.6 (Spatial images of graph ends)**.** *Let $K_n \uparrow \mathscr{H}$ be a finite graph exhaustion and let $C_n(\omega)$ be the nested components representing an end ω. In the Riemann sphere $\widehat{\mathbb{C}}$,*

$$\bigcap_{n \geq 1} \overline{\bigcup_{H \in C_n(\omega)} H}^{\widehat{\mathbb{C}}} = \{p_\omega\} \tag{24}$$

for a unique $p_\omega \in S \cup \{\infty\}$. Every spherical neighborhood of p_ω contains the entire tail-cell union for all sufficiently large n. The point p_ω is independent of the exhaustion, of the admissible witness, and of the cell representatives. The map $\omega \mapsto p_\omega$ is continuous in the usual end topology. Moreover

$$p_\omega = \infty \quad \Longleftrightarrow \quad \inf_{H \in C_n(\omega)} \operatorname{dist}(0, H) \longrightarrow \infty. \tag{25}$$

Proof. The spherical closure F_n of the tail-cell union is nonempty, compact, connected, and nested: graph connectivity plus adjacency-implies-intersection makes each union connected. Hence $F = \cap_n F_n$ is nonempty compact and connected. If $z \notin S$, a neighborhood of z meets only finitely many cells; once all of them lie in K_n, a smaller neighborhood misses F_n. Thus $F \subset S \cup \{\infty\}$.

The latter set has zero one-dimensional Hausdorff measure in the chordal metric. A connected metric set containing two points a positive distance apart has positive one-dimensional Hausdorff measure, because the distance from one of the points maps it continuously onto an interval. Hence F is a singleton. Nested compactness then gives eventual containment of F_n in every neighborhood of the singleton. Cofinality gives exhaustion independence; the formula itself uses only cells and adjacency, so the image is witness-independent. The end-topology continuity and (25) follow immediately from eventual tail containment. □

4 A rooted-partition projection lemma

Both the spatial corrector argument and the temporal ergodic argument use the same measure-theoretic fact. Stating it once removes the need to describe an environment-dependent "invariant sigma-field" informally.

Lemma 4.1 (Class-kernel conditional expectation)**.** *Let $(\Omega, \mathcal{A}, \mu)$ be a probability space and let K be a Markov kernel on $\mathcal{A}$. Assume:*

(i) $\mu K = \mu$;

(ii) for the measure $M(d\omega, d\eta) = \mu(d\omega)K(\omega, d\eta)$,

$$K(\eta, \cdot) = K(\omega, \cdot) \qquad \text{for } M\text{-a.e. } (\omega, \eta). \tag{26}$$

Define

$$\mathcal{G} = \{A \in \mathcal{A} : \mathbf{1}_A(\eta) = \mathbf{1}_A(\omega) \text{ for } M\text{-a.e. } (\omega, \eta)\}, \tag{27}$$

completed under μ. *Then* $\mathcal{G}$ *is a sigma-field and, for every* $F \in L^1(\mu)$,

$$KF = \mathbb{E}_\mu[F \mid \mathcal{G}]. \tag{28}$$

If K_m *is a family of such kernels whose classes are nested so that every* K_n*-class is a union of* K_m*-classes for* $n \geq m$, *then* $\mathcal{G}_n \subset \mathcal{G}_m$ *and* $(K_m F)$ *is a reverse martingale.*

Proof. The class-saturated events in (27) are closed under complements and countable unions, so they form a sigma-field modulo null sets. By (26), KF is constant along K-classes and therefore $\mathcal{G}$-measurable. If $A \in \mathcal{G}$, then $\mathbf{1}_A(\omega) = \mathbf{1}_A(\eta)$ for M-a.e. pair, hence

$$\int_A KF \, d\mu = \int \mu(d\omega) \int K(\omega, d\eta)\mathbf{1}_A(\omega)F(\eta) = \int \mu(d\omega) \int K(\omega, d\eta)\mathbf{1}_A(\eta)F(\eta).$$

Using $\mu K = \mu$, the last expression equals $\int_A F \, d\mu$. This is the defining property of conditional expectation. The nesting statement is immediate from the class description. □

The point of Lemma 4.1 is that no group action, quotient space, or standard-Borel theorem is needed. One only checks that rerooting uniformly inside the current block preserves the relevant law on the scale-invariant sigma-field and that rerooting twice does not change the block-class kernel.

5 Spatial estimates and dyadic blocks

5.1 No macroscopic cells and energy density

Lemma 5.1 (Spatial finiteness above a scale)**.** *Almost surely, for every* $K > 0$,

$$\#\{H : \operatorname{dist}(0, H) \leq K d_H\} < \infty. \tag{29}$$

Consequently, for every finite R *and every* $\delta > 0$, *only finitely many cells meeting* $B_R(0)$ *have diameter at least* δ, *and*

$$D_R := \sup_{H \in \mathcal{H}(B_R(0))} d_H < \infty, \qquad \frac{D_R}{R} \longrightarrow 0. \tag{30}$$

Proof. Every cell has a neighbor, so $\pi(H) + \pi^*(H) \geq 2$ and (FE) gives $\mathbb{E}[d_{H_0}^2 / a_{H_0}] < \infty$. For H let

$$H^{[K]} = \{z : \operatorname{dist}(z, H) \leq K d_H\}.$$

The transport

$$T_K(\mathcal{H}, w, z) = a_{H_w}^{-1} \mathbf{1}_{\{z \in H_w^{[K]}\}}$$

has expected outgoing mass at most $\pi(K+1)^2 \mathbb{E}[d_{H_0}^2 / a_{H_0}]$, while the incoming mass counts the cells in (29). Apply mass transport and intersect over integer K.

A cell meeting B_R with $d_H \geq \delta$ is counted in (29) for $K \geq R/\delta$. Taking $\delta = 1$ gives $D_R < \infty$. If $d_H \geq \varepsilon R$ and H meets B_R, then it belongs to the fixed finite set counted with $K \geq 1/\varepsilon$; for large R none of those finitely many fixed diameters exceeds εR. □

We use an independent uniform dyadic system $\mathcal{D}$ as in GMS: its side lengths are 2^{s+k}, s is uniform modulo one, and successive parents are chosen uniformly. The law is invariant under deterministic translations and positive dilations.

For a dyadic square Q, let $Q^{(j)}$ be its jth ancestor and define

$$D(Q) = \sup_{H \cap Q \neq \varnothing} d_H, \quad a(Q) = \sup_{j \geq 0} \frac{D(Q^{(j)})}{\ell(Q^{(j)})}, \quad b(Q) = a(Q)^{-1}, \quad \kappa(Q) = \sum_{j \geq 0} 4^{-j} b(Q^{(j)}). \tag{31}$$

By (30), all these quantities are finite and positive. If $P = Q^{(1)}$,

$$b(Q) \leq b(P) \leq 2b(Q), \qquad \frac{4}{3} b(Q) < \kappa(Q) \leq 2b(Q), \qquad \kappa(P) = 4(\kappa(Q) - b(Q)) > \kappa(Q). \tag{32}$$

Thus κ tends to infinity along ancestors and to zero down a chain through a point of a cell.

For $m > 0$ let

$$\mathcal{S}_m = \{Q \in \mathcal{D} : \kappa(Q) \leq m < \kappa(Q^{(1)})\}. \tag{33}$$

These squares have disjoint interiors and their closures cover every point belonging to a cell. They coarsen as m increases and commute with similarities.

5.2 Spatial block averaging without joint grid ergodicity

Let ω denote the environment together with any independent covariant marks, and let $\mathcal{D}'$ be a further independent uniform dyadic system used only for averaging. Write $\widehat{\omega} = (\omega, \mathcal{D}')$ and let $\widehat{\mathbb{P}}$ be the product law. For a scale-invariant event or functional, work on the scale-invariant sigma-field $\mathcal{A}_{\mathrm{sc}}$ of this marked state. For fixed m, define the rerooting kernel

$$K_m(\widehat{\omega}, A) = \frac{1}{\ell(S_m(0))^2} \int_{S_m(0)} \mathbf{1}_A(\omega - z, \mathcal{D}' - z)\, dz, \qquad A \in \mathcal{A}_{\mathrm{sc}}, \tag{34}$$

where $S_m(0)$ is the selected square containing the origin off the null union of selected boundaries.

Proposition 5.2 (Spatial block projection)**.** *For every m, K_m satisfies the hypotheses of Lemma 4.1 on the marked probability space $(\widehat{\Omega}, \mathcal{A}_{\mathrm{sc}}, \widehat{\mathbb{P}})$. Hence for every integrable scale-invariant F,*

$$A_m := \frac{1}{\ell(S_m(0))^2} \int_{S_m(0)} F(\omega - z)\, dz = \mathbb{E}[F \mid \mathcal{G}_m]. \tag{35}$$

The sigma-fields $\mathcal{G}_m$ decrease as m increases.

Proof. If z lies in the interior of $S_m(0)$, covariance of the selected partition gives

$$S_m^{\omega - z}(0) = S_m^{\omega}(0) - z.$$

Therefore rerooting anywhere in the block gives the same uniform rerooting kernel; this is the class-kernel property (26). To prove stationarity on $\mathcal{A}_{\mathrm{sc}}$, let $U \geq 0$ be scale invariant and use the marked mass transport

$$T(\omega, w, z) = \frac{\mathbf{1}_{\{z \in S_m(w)\}}}{\ell(S_m(w))^2} U(\omega - z, \mathcal{D}' - z).$$

It has scaling degree -2. Its incoming integral is $U(\omega, \mathcal{D}')$, while its outgoing integral is $K_m U$. Averaging over the independent marks before applying (6) gives $\mathbb{E}[K_m U] = \mathbb{E}[U]$. Thus $\widehat{\mathbb{P}} K_m = \widehat{\mathbb{P}}$ on the scale-invariant sigma-field. Lemma 4.1 gives (35). The selected partitions coarsen with m, so the class sigma-fields decrease. □

Proposition 5.3 (Spatial maximal inequality)**.** *If $F \geq 0$ is integrable and scale invariant and $\rho_\omega(z) = F(\omega - z)$, then ρ is locally integrable almost surely and*

$$\mathbb{P}\left[\sup_{R>0} \frac{1}{R^2} \int_{B_R(0)} \rho(z)\, dz > \lambda\right] \leq \frac{512\mathbb{E}[F]}{\lambda}. \tag{36}$$

No joint ergodicity of ω and the auxiliary grid is assumed.

Proof. For any finite increasing list of positive rational m's, (35) read in reverse order is a nonnegative martingale. Doob's weak-L^1 inequality and exhaustion of the positive rationals give

$$\mathbb{P}\left[\sup_{m\in\mathbb{Q}_{>0}} A_m > t\right] \leq \frac{\mathbb{E}[F]}{t}.$$

Every origin dyadic square appears as $S_m(0)$ for an interval of m containing a rational, so this controls all origin dyadic averages. These squares exhaust the plane, proving local integrability.

On the event that the centered-ball maximal function exceeds λ, choose a positive rational witness radius R. Conditional on the environment, choose a dyadic level with side length in $[8R, 16R)$. With probability at least $(3/4)^2 > 1/2$ the origin square contains $B_R(0)$, and on that event its average exceeds $\lambda/256$. Integrate over the independent grid and apply the dyadic maximal bound. □

Corollary 5.4 (Spatial energy bounds)**.** *Almost surely,*

$$W(R) := \sum_{H\in\mathcal{H}(B_R(0))} d_H^2\big(\pi(H) + \pi^*(H)\big) < \infty, \qquad \limsup_{R\to\infty} \frac{W(R)}{R^2} < \infty. \tag{37}$$

Under environment ergodicity modulo similarities, the last limsup is deterministic.

Proof. Apply Proposition 5.3 to

$$F_q(\mathcal{H}) = \frac{d_{H_0}^2}{a_{H_0}}\big(\pi(H_0) + \pi^*(H_0)\big).$$

Every cell meeting B_R is contained in B_{R+D_R}, so integrating the rooted density counts its whole weight and gives

$$W(R) \leq (R + D_R)^2 M(\rho_q).$$

Use (30). The limsup is unchanged by translating the origin and scales quadratically under a common dilation, hence its normalized value is similarity invariant. □

6 Full-energy harmonic coordinates

6.1 The full finite-energy Dirichlet problem

Definition 6.1. *For a countable graph G and $A \subset V(G)$, say that (G, A) is* boundary anchored *if every connected component of G contains at least one vertex of A.*

Lemma 6.2 (Full-energy Dirichlet problem)**.** *Let G be a countable locally finite conductance graph, not necessarily connected, and let (G, A) be boundary anchored. If $g : V(G) \to \mathbb{R}^k$ has finite energy, then there is a unique minimizer h of $\mathcal{E}_G(f)$ among all f with $f|_A = g|_A$. It is characterized by*

$$\mathcal{E}_G(h, u) = 0 \qquad \text{for every finite-energy } u \text{ with } u|_A = 0. \tag{38}$$

Coordinatewise bounds on the boundary trace propagate to all vertices in the corresponding components.

Proof. Orient the edges and use the Hilbert space with norm $\sum_e c(e)|\theta(e)|^2$. On each component choose one anchor in A. The gradients of finite-energy functions vanishing on A form a closed subspace: if du_n is Cauchy, then for every vertex x, a fixed finite path from an anchor a to x gives

$$|u_n(x) - u_m(x)|^2 \le \left(\sum_{e \in P_{a,x}} c(e)^{-1} \right) \mathcal{E}_G(u_n - u_m),$$

so the potentials converge pointwise to a zero-trace finite-energy function. Project dg onto this closed subspace and subtract the corresponding anchored potential. Orthogonality gives (38) and minimality. Uniqueness follows because a zero-energy difference is constant on each component and vanishes at an anchor. Clipping one coordinate cannot increase energy, so the maximum principle follows from uniqueness. □

Lemma 6.3 (Spatial restrictions are boundary anchored)**.** *For every bounded closed axis-parallel rectangle Q, (G_Q, A_Q) is boundary anchored. If $u \in \mathcal{V}_Q$, extension by zero to the full graph has exactly the energy $\mathcal{E}_{G_Q}(u)$. Moreover $b|_{G_Q}$ has finite energy.*

Proof. A cell meeting Q but not ∂Q is contained in Q°. Every neighbor of such a cell meets Q because adjacent cells intersect. Hence every edge from G_Q to its complement has its G_Q endpoint in A_Q. A component of G_Q disjoint from A_Q would therefore have no edge to its complement in the globally connected graph, forcing every cell to lie in the bounded rectangle, contradicting almost-everywhere coverage of the plane. This proves anchoring and the zero-extension assertion.

By (4) and Corollary 5.4,

$$\mathcal{E}_{G_Q}(b) \le 2 \sum_{H \in \mathcal{H}(Q)} d_H^2 \pi(H) < \infty.$$

□

The minimizers in Lemma 6.2 are measurable in the environment. Indeed, enumerate the vertices by Proposition 2.8; exhaust each component by finite induced subgraphs containing a chosen finite path to its first boundary anchor; solve the corresponding finite Dirichlet systems; and take the pointwise limit. This finite approximation also shows convergence of the gradients in the edge Hilbert space. We use this only for measurability, not to impose a finite-support condition at an accumulation point.

6.2 Blockwise minimizers and specific-energy projection

Fix one independent uniform dyadic system $\mathcal{D}$. For $m \in \mathbb{N}$, let $\mathcal{S}_m$ be (33) and let

$$\mathcal{B}_m = \{H : H \cap \bigcup_{Q \in \mathcal{S}_m} \partial Q \neq \varnothing\}.$$

Set $\phi_0(H) = b_H$. On each $Q \in \mathcal{S}_m$, define ϕ_m to be the full-energy minimizer on G_Q with boundary trace b on A_Q. The definitions agree on overlaps because a cell meeting two selected blocks is pinned to b_H. A cell not in $\mathcal{B}_m$ lies in the interior of exactly one selected square. Thus ϕ_m is a globally defined measurable function.

The selected partitions coarsen, so for $n \ge m$,

$$\mathcal{B}_n \subset \mathcal{B}_m, \qquad \phi_n = \phi_m = \phi_0 \quad \text{on } \mathcal{B}_n. \tag{39}$$

The construction is exactly covariant under translations and positive dilations, with the potential itself scaling linearly.

For a covariant edge field θ whose values scale linearly, define

$$\|\theta\|_{\mathrm{sp}}^2 = \mathbb{E}\left[\frac{1}{2a_{H_0}} \sum_{K\sim H_0} c(H_0,K)|\theta(H_0,K)|^2\right]. \tag{40}$$

Let $g_m = d\phi_m$ and $e_m = \|g_m\|_{\mathrm{sp}}^2$.

Lemma 6.4 (Finite base energy)**.** *$e_0 < \infty$; more precisely,*

$$e_0 \le 2\mathbb{E}\left[\frac{d_{H_0}^2\pi(H_0)}{a_{H_0}}\right].$$

Proof. By (4), the rooted density is bounded by

$$\frac{d_{H_0}^2\pi(H_0)}{a_{H_0}} + \frac{1}{a_{H_0}} \sum_{K\sim H_0} c(H_0,K) d_K^2.$$

The two terms have equal expectation by mass transport, sending the K-endpoint contribution across each edge. □

For a selected square Q, call an edge *active in Q* if both endpoints lie in G_Q and at least one endpoint is not in A_Q. Active edge sets of distinct selected squares are disjoint. Every edge on which a variation vanishing on $\mathcal{B}_m$ has nonzero gradient is active in exactly one selected square.

Lemma 6.5 (Block redistribution)**.** *Suppose $q_e \ge 0$ is a covariant edge coefficient, scales quadratically, and is zero unless e is active. Let Q_e be the unique block owning e. Then*

$$\mathbb{E}\left[\frac{1}{2a_{H_0}} \sum_{e\ni H_0} q_e\right] = \mathbb{E}\left[\frac{1}{\ell(S_m(0))^2} \sum_{e:Q_e=S_m(0)} q_e\right]. \tag{41}$$

The same identity holds for signed coefficients when the rooted density of $|q_e|$ is integrable.

Proof. For $e = \{H, K\}$ send its endpoint-spread mass uniformly to its owner block:

$$T(\omega,w,z) = \sum_e \frac{q_e}{2}\left(\frac{\mathbf{1}_{\{w\in H\}}}{a_H} + \frac{\mathbf{1}_{\{w\in K\}}}{a_K}\right)\frac{\mathbf{1}_{\{z\in Q_e\}}}{\ell(Q_e)^2}.$$

The transport has scaling degree -2; its outgoing and incoming integrals are the two sides of (41). Apply marked mass transport and Tonelli. For signed coefficients use positive and negative parts. □

Proposition 6.6 (Nested energy projections)**.** *For every m,*

$$e_m + \|g_m - g_0\|_{\mathrm{sp}}^2 = e_0. \tag{42}$$

For $n \ge m$,

$$e_m = e_n + \|g_m - g_n\|_{\mathrm{sp}}^2. \tag{43}$$

Hence $e_m \downarrow e_\infty$ and (g_m) is Cauchy in specific energy.

Proof. On one m-block Q, put $v = \phi_m - b$. Full variational orthogonality gives $\mathcal{E}_{G_Q}(\phi_m, v) = 0$. Only active edges contribute to dv, so Cauchy–Schwarz gives

$$\mathcal{E}_{G_Q}(v) \le \sum_{e \text{ active in } Q} c(e)|g_0(e)|^2.$$

Redistribution implies $\|g_m - g_0\|_{\mathrm{sp}} < \infty$. Apply Lemma 6.5 to the signed coefficients

$$c(e)g_m(e) \cdot (g_m(e) - g_0(e)).$$

Their block sums vanish by orthogonality, hence $\langle g_m, g_m - g_0\rangle_{\mathrm{sp}} = 0$, which is (42).

If $n \ge m$, $\phi_m - \phi_n$ vanishes on $\mathcal{B}_n$ by (39). Orthogonality of the n-block minimizers and the same signed redistribution give $\langle g_n, g_m - g_n\rangle_{\mathrm{sp}} = 0$, proving (43). □

6.3 Limit and local harmonicity

For a covariant finite-specific-energy edge field θ, let ρ_θ denote its rooted density in (40). Proposition 5.3 gives

$$\sum_{e \in E(\mathcal{H}(B_R(0)))} c(e)|\theta(e)|^2 \le \int_{B_{R+D_R}(0)} \rho_\theta(z)\, dz \le (R + D_R)^2 M(\rho_\theta) \tag{44}$$

almost surely. In particular a zero specific norm forces the field to vanish on every edge.

Proposition 6.7 (Existence and uniqueness of the limiting gradient)**.** *There is a measurable covariant exact gradient $g = d\Phi$ such that*

$$\|g\|_{\mathrm{sp}}^2 = e_\infty, \qquad \|g_m - g\|_{\mathrm{sp}}^2 = e_m - e_\infty \longrightarrow 0. \tag{45}$$

After normalization $\Phi(H_0) = 0$, a deterministic subsequence converges almost surely in energy on every bounded spatial patch and pointwise at every vertex. Moreover (11) holds on every bounded rectangle.

Proof. Choose deterministic $m_j \uparrow \infty$ with $\|g_{m_{j+1}} - g_{m_j}\|_{\mathrm{sp}}^2 \le 2^{-4j}$. By Proposition 5.3 and Borel–Cantelli, the maximal rooted density of the successive difference is at most 2^{-2j} for all large j. Equation (44) then makes the square roots of the successive patch energies summable on every bounded patch. Thus the gradients converge strongly on each patch. Every finite graph cycle is eventually contained in one patch, so the limiting cycle sum is zero. Global connectedness permits path integration from H_0, defining Φ.

Fatou and (43) give one inequality in (45); passing to the limit in the Pythagorean identity gives equality. Fix a rectangle Q. For all sufficiently large m, the entire cell union meeting Q lies in the interior of one selected ancestor block. Every $u \in \mathcal{V}_Q$, extended by zero as in Lemma 6.3, is therefore an admissible variation for ϕ_m. Strong patch convergence yields $\mathcal{E}_{G_Q}(\Phi, u) = 0$. No uncountable intersection of exceptional events is needed because the local-convergence event is fixed on integer-radius patches. □

Proposition 6.8 (The auxiliary grid disappears)**.** *The gradient g and normalized potential Φ in Proposition 6.7 are measurable functions of the unmarked environment alone.*

Proof. Condition on $\mathcal{H}$ and take two independent grids $\mathcal{D}_1, \mathcal{D}_2$, producing g^1, g^2. The limit Φ^1 is orthogonal to every full finite-energy variation vanishing on the boundary of a block of the second grid. Applying signed redistribution and then sending the second block parameter to infinity gives

$$\langle g^1, g^2 - g_0\rangle_{\mathrm{sp}} = 0.$$

The same relation with the first grid gives $\langle g^1, g^1 - g_0\rangle_{\mathrm{sp}} = 0$, hence $\langle g^1, g^1 - g^2\rangle_{\mathrm{sp}} = 0$. Interchanging the grids and adding yields $\|g^1 - g^2\|_{\mathrm{sp}} = 0$. Equation (44) then gives equality on every edge almost surely.

Thus, conditional on $\mathcal{H}$, two independent copies of each bounded transform of every countable edge coordinate agree almost surely. The conditional variance is zero, so the coordinates are $\mathcal{H}$-measurable. Path integration yields an unmarked measurable Φ. Exact similarity covariance is obtained by defining the limit on the similarity-invariant full-probability convergence set and using zero off that set. □

6.4 Uniform sublinearity

Lemma 6.9 (Fixed blocks are submacroscopic)**.** *For every fixed m,*

$$s_m(R) := \sup\{\ell(Q) : Q \in \mathcal{S}_m,\ Q \cap B_R(0) \neq \varnothing\} \quad \textit{satisfies} \quad \frac{s_m(R)}{R} \to 0.$$

Consequently

$$\frac{1}{R} \sup_{H \in \mathcal{H}(B_R(0))} |\phi_m(H) - b_H| \longrightarrow 0. \tag{46}$$

Proof. Fix $\varepsilon > 0$. If Q meets B_R and $\ell(Q) \geq \varepsilon R$, every ancestor T of Q lies in a disk of radius at most $(\varepsilon^{-1} + \sqrt{2})\ell(T)$. Equation (30) therefore makes $D(T)/\ell(T) \to 0$ uniformly over all such ancestors as $R \to \infty$. Hence $a(Q) \to 0$ and $\kappa(Q) \to \infty$, contradicting $\kappa(Q) \leq m$. This proves the first assertion.

If a cell is pinned, $\phi_m = b$. Otherwise it lies in the interior of one selected block Q meeting B_R. The coordinatewise maximum principle bounds ϕ_m by the centroids of the boundary cells of Q. Their displacement from Q is at most $D_{2R} = o(R)$, while $\ell(Q) = o(R)$ uniformly. This yields (46). □

Let $r_m = \Phi - \phi_m$. By (45), choose a deterministic subsequence n_j with $\|dr_{n_j}\|_{\mathrm{sp}}^2 \leq 2^{-4j}$. Then almost surely

$$\eta_j := \limsup_{R \to \infty} R^{-2} \mathcal{E}_{\mathcal{H}(B_R(0))}(r_{n_j}) \longrightarrow 0. \tag{47}$$

Lemma 6.10 (Energy controls line variation)**.** *Let $Q_R = [-2R, 2R]^2$ and $L_y = [-2R, 2R] \times \{y\}$. For a finite-energy function f on G_{Q_R} define*

$$V_f(y) = \sum_{\{H,K\} \in E(\mathcal{H}(L_y))} |f(K) - f(H)|.$$

Then

$$\int_{-2R}^{2R} V_f(y)\, dy \leq \left(\sum_{H \in \mathcal{H}(Q_R)} d_H^2 \pi^*(H) \right)^{1/2} \mathcal{E}_{G_{Q_R}}(f)^{1/2}. \tag{48}$$

The same holds for vertical lines. If $y \notin N_h$, then $\operatorname{osc}_{\mathcal{H}(L_y)} f \leq V_f(y)$ whenever the latter is finite.

Proof. For an edge $e = \{H, K\}$ let λ_e be the length of the set of y for which both endpoints meet L_y. Then $\lambda_e \leq \min(d_H, d_K)$. Tonelli and Cauchy–Schwarz give

$$\int V_f \leq \left(\sum_e \frac{\lambda_e^2}{c(e)} \right)^{1/2} \mathcal{E}_{G_{Q_R}}(f)^{1/2} \leq \left(\sum_{H \in \mathcal{H}(Q_R)} d_H^2 \pi^*(H) \right)^{1/2} \mathcal{E}_{G_{Q_R}}(f)^{1/2}.$$

For a good horizontal offset, Proposition 3.3 joins any two cells meeting the segment by a finite path inside the induced graph, so oscillation is bounded by the total edge variation. □

Proposition 6.11 (Uniform sublinearity)**.** *Equation* (13) *holds almost surely.*

Proof. Work on the common full-probability event for all preceding estimates. By (37), for all large R,

$$\sum_{H\in\mathcal{H}(Q_R)} d_H^2\pi^*(H) \le KR^2$$

for a finite random K. Combining this with (47) and Lemma 6.10 gives

$$\lim_{j\to\infty}\limsup_{R\to\infty} R^{-2}\int_{-2R}^{2R} V_{r_{n_j}}(y)\,dy = 0, \tag{49}$$

and likewise vertically.

Fix $N \ge 5$ and put $\alpha = 1/N$. Choose one fixed $m = n_j$ so large that, for all sufficiently large R, both line-variation integrals are at most $\alpha^2R^2/4$. Markov's inequality shows that the set of horizontal offsets with $V_{r_m}(y) > \alpha R$ has length at most $\alpha R/4$, and similarly vertically. Partition $[-2R, 2R]$ into intervals of length αR. Each interval contains a horizontal offset outside both the variation-bad set and N_h, and a vertical offset outside both the variation-bad set and N_v. Choose one of each. The resulting grid has mesh at most $2\alpha R$ and surrounds B_R with positive margin.

Let T_R be the union of cells meeting the selected grid lines. Each selected line has residual oscillation at most αR, and a selected horizontal and vertical line meet at a covered point outside S. Connecting through at most three lines gives

$$\operatorname{osc}_{T_R} r_m \le 3\alpha R.$$

By Lemma 6.9, $\chi_m := \phi_m - b$ is $o(R)$ uniformly on Q_R. Hence, for $\chi = \Phi - b$,

$$\operatorname{osc}_{T_R} \chi \le 3\alpha R + o(R). \tag{50}$$

Choose $H_* \in T_R$ and let $c_R = \chi(H_*)$. If H meets B_R but not T_R, connectedness of H places it in the interior of one grid rectangle Q', whose side lengths are at most $2\alpha R$. Every boundary cell of Q' lies in T_R, and

$$|b_H - b_{H'}| \le \operatorname{diam}(Q') + d_H + d_{H'} \le 2\sqrt{2}\alpha R + 2D_{3R}.$$

Apply the full-energy maximum principle to Φ on $G_{Q'}$ with boundary values controlled by (50). It follows, uniformly for $H \in \mathcal{H}(B_R)$, that

$$|\chi(H) - c_R| \le 12\alpha R + o(R).$$

The same bound holds for H_0, so

$$\sup_{H\in\mathcal{H}(B_R)} |\chi(H) - \chi(H_0)| \le 24\alpha R + o(R).$$

Since $\chi(H_0) = -b_{H_0}$ is fixed, the limsup in (13) is at most $24/N$. Send $N \to \infty$. If $z_H \in H$, then $|z_H - b_H| \le d_H$ and (30) gives the representative-independent version. □

Theorem 1.4 follows from Propositions 6.7, 6.8, and 6.11.

7 Excluding spatial infinity

This section is deterministic. Let $G = (V, E, c)$ be a connected locally finite network and let Y be the canonical reflected process of Proposition 1.6 with an admissible fast clock. Fix a starting vertex o. Put

$$\sigma_o = \inf\{t > 0 : Y_t \neq o\}, \qquad \tau_o^+ = \inf\{t \geq \sigma_o : Y_t = o\}.$$

These times are finite and attained at vertices.

For the spatial application, $V = \mathscr{H}$, choose $z_H \in H$, and assume that for some finite C, R_*,

$$D(R) := \sup_{H \cap B_R(0) \neq \varnothing} d_H = o(R), \qquad W_\pi(R) := \sum_{H \cap B_R(0) \neq \varnothing} d_H^2 \pi(H) \leq CR^2 \tag{51}$$

for $R \geq R_*$. Lemma 5.1 and Corollary 5.4 give these bounds almost surely.

7.1 Finite-target excursion estimate

Lemma 7.1 (Finite-target minimizer)**.** *Let $F \subset V \setminus \{o\}$ be nonempty and finite. There is a unique finite-energy minimizer h_F among functions satisfying*

$$h_F(o) = 0, \qquad h_F|_F = 1.$$

It takes values in $[0, 1]$ and

$$\mathcal{E}_G(h_F, \eta) = 0 \tag{52}$$

for every finite-energy η vanishing on $\{o\} \cup F$.

Proof. The competitor $1 - \mathbf{1}_{\{o\}}$ has finite energy $\pi(o)$. Clip a minimizing sequence to $[0, 1]$ and use diagonal compactness on the countable vertex set; Fatou gives a minimizer. Strict convexity modulo constants and the value at o give uniqueness. Varying by $t\eta$ yields (52). □

Lemma 7.2 (Reflected excursion identity)**.** *Let $\tau_F = \inf\{t \geq 0 : Y_t \in F\}$. Then*

$$\mathbb{P}_o^{\mathrm{ref}}[\tau_F < \tau_o^+] = \frac{\mathcal{E}_G(h_F)}{\pi(o)}. \tag{53}$$

Consequently, if f has finite energy, $f(o) = 0$, and $f|_F = 1$, then

$$\mathbb{P}_o^{\mathrm{ref}}[\tau_F < \tau_o^+] \leq \frac{\mathcal{E}_G(f)}{\pi(o)}. \tag{54}$$

Proof. By Proposition 1.6(ii), $h_F(x)$ is the probability that the reflected process started from x first hits F rather than o. At the first departure from o, the neighbor x is chosen with probability $c(o, x)/\pi(o)$. Hence

$$\mathbb{P}_o^{\mathrm{ref}}[\tau_F < \tau_o^+] = \frac{1}{\pi(o)} \sum_{x \sim o} c(o, x) h_F(x).$$

Let $u = 1 - \mathbf{1}_{\{o\}}$. Since $u - h_F$ vanishes on $\{o\} \cup F$, (52) gives

$$\mathcal{E}_G(h_F) = \mathcal{E}_G(h_F, u) = \sum_{x \sim o} c(o, x) h_F(x).$$

This proves (53); minimality gives (54). □

By monotone convergence, the same upper bound holds when the target F is replaced by an arbitrary countable set D on which $f = 1$.

Proposition 7.3 (Vanishing-energy cutoff criterion)**.** *Let $\rho : V \to [0, \infty)$ be finite-valued. Suppose there are $R_n \uparrow \infty$ and finite-energy $f_n : V \to [0, 1]$ such that*

$$f_n(o) = 0, \qquad f_n(x) = 1 \text{ when } \rho(x) \geq R_n, \qquad \mathcal{E}_G(f_n) \to 0.$$

Then almost surely the ρ-values at vertex-valued times are bounded on every bounded time interval. Consequently no reflected time has an end label ω for which

$$\inf_{x \in C_m(\omega)} \rho(x) \longrightarrow \infty \tag{55}$$

for a finite graph exhaustion.

Proof. During one excursion from o to its next return, the probability of visiting $\{\rho \geq R_n\}$ is at most $\mathcal{E}_G(f_n)/\pi(o)$. Thus the vertex-valued ρ-range of that excursion is almost surely bounded. Successive return times tend to infinity because the holding intervals at o have independent positive exponential lengths. Hence a bounded time interval intersects only finitely many excursions, proving local boundedness.

If at time t the end label satisfied (55), choose a bounded time interval containing t and let M bound the vertex-valued ρ-range there. An end neighborhood $C_m(\omega)$ lies inside $\{\rho > M + 1\}$. The local end-label property in Proposition 1.6(iii) forces all sufficiently close vertex-valued rational times into this neighborhood, contradicting the bound M. □

7.2 Logarithmic spatial cutoffs

Choose $r_0 > |z_o|$ so large that $r_0 \geq R_*$ and $D(R) \leq R/100$ for $R \geq r_0$. Set

$$s_H = \max\{r_0, |z_H|\}, \qquad f_n(H) = \min\left\{1, \frac{\log(s_H/r_0)}{n \log 2}\right\}. \tag{56}$$

Proposition 7.4 (Annular energy estimate)**.** *$f_n(o) = 0$, $f_n(H) = 1$ for $|z_H| \geq 2^n r_0$, and*

$$\mathcal{E}_G(f_n) \leq \frac{32C}{n(\log 2)^2}. \tag{57}$$

Proof. Since $z_H \in H$ and H meets $B_{s_H}(0)$,

$$d_H \leq s_H/100. \tag{58}$$

If $H \sim K$, choose a point in $H \cap K$. Then

$$|z_H - z_K| \leq d_H + d_K, \qquad |s_H - s_K| \leq d_H + d_K \leq (s_H + s_K)/100,$$

so $\max(s_H, s_K)/\min(s_H, s_K) < 2$.

For $j = 0, \ldots, n-1$, let E_j be the edges on which f_n is nonconstant and

$$2^j r_0 \leq \min(s_H, s_K) < 2^{j+1} r_0.$$

These sets partition the nonconstant edges and both endpoint representatives of an edge in E_j lie in $B_{2^{j+2} r_0}(0)$. The clipped logarithm is $[n(\log 2) 2^j r_0]^{-1}$-Lipschitz on this annulus, hence

$$|f_n(H) - f_n(K)| \leq \frac{d_H + d_K}{n(\log 2) 2^j r_0}.$$

Therefore

$$\sum_{\{H,K\}\in E_j} c(H,K)|df_n|^2 \le \frac{2W_\pi(2^{j+2}r_0)}{n^2(\log 2)^2 4^j r_0^2} \le \frac{32C}{n^2(\log 2)^2}.$$

Summing over j proves (57). □

Theorem 7.5 (Spatial boundedness of the reflected path)**.** *Under* (51)*, for every starting cell o and every admissible reflected clock,*

$$\mathbb{P}_o^{\mathrm{ref}}\left[\exists t \in [\sigma_o, \tau_o^+) : |z_{Y_t}| \ge 2^n r_0\right] \le \frac{32C}{\pi(o)(\log 2)^2 n}. \tag{59}$$

Almost surely,

$$\sup\{|z_{Y_t}| : 0 \le t \le T,\ Y_t \in V\} < \infty \qquad \forall T < \infty, \tag{60}$$

and no reflected time has an end at spatial infinity.

Proof. Apply Lemma 7.2 and Proposition 7.4 to the target $\{|z_H| \ge 2^n r_0\}$. Since the cutoff energies tend to zero, Proposition 7.3 gives (60) and simultaneous avoidance of all ends satisfying (55) with $\rho(H) = |z_H|$. By (58), this end condition is equivalent to (25), hence to $p_\omega = \infty$ by Proposition 3.6. □

Combining Theorem 7.5 with Proposition 3.6, every end-valued time of the reflected path has a Euclidean spatial limit in the admissible singular set, simultaneously for all such times and independently of the representatives.

8 The area and exact clocks

Fix an environment in the common full-probability set supplied by the preceding sections. Choose an admissible fast rate w_0 so large that

$$m_0(H) = \frac{\pi(H)}{w_0(H)} > 0, \qquad \sum_H m_0(H) < \infty. \tag{61}$$

This is possible by increasing the rate of the jth labeled vertex so that $m_0(H_j) \le 2^{-j}$. Let Y be the corresponding recurrent reflected process. By Proposition 1.6(iv), its symmetric form on $L^2(m_0)$ has full finite-energy extended space and graph energy (8).

8.1 Occupation of a return cycle

Lemma 8.1 (Return-cycle occupation)**.** *For a starting vertex o and every nonnegative $f : V \to [0,\infty]$,*

$$\mathbb{E}_o \int_0^{\tau_o^+} f(Y_s)\, ds = \frac{1}{\pi(o)} \sum_{H\in V} m_0(H) f(H). \tag{62}$$

Both sides may be infinite. In particular, if N_H is the number of holding intervals at H in one complete return cycle (including the initial interval when $H = o$), then

$$\mathbb{E}_o N_H = \frac{\pi(H)}{\pi(o)}. \tag{63}$$

Proof. For $H = o$, the expected occupation before the initial departure is $m_0(o)/\pi(o)$. For $x, y \neq o$, let

$$K_o(x,y) = \frac{1}{m_0(y)} \mathbb{E}_x \int_0^{\tau_o} \mathbf{1}_{\{Y_s = y\}}\, ds.$$

The process killed on hitting o is symmetric, hence $K_o(x,y) = K_o(y,x)$. Since every hit of o arrives by an ordinary jump from one of its finitely many neighbors, compensation of these vertex jumps gives

$$\sum_{x \sim o} c(o,x) K_o(y,x) = 1.$$

For completeness, first stop at a deterministic fast-time horizon, compensate only the finitely many possible jumps from neighbors of o, and then let the horizon tend to infinity. Monotone convergence gives the displayed identity. No compensation at a graph end is used. After the initial jump from o, the Markov property and symmetry therefore give, for $y \neq o$,

$$\mathbb{E}_o \int_0^{\tau_o^+} \mathbf{1}_{\{Y_s = y\}}\, ds = \frac{m_0(y)}{\pi(o)} \sum_{x \sim o} c(o,x) K_o(x,y) = \frac{m_0(y)}{\pi(o)}.$$

Sum with weights $f(y)$ and use Tonelli. Finally, each arrival to H contributes a holding interval of conditional mean $m_0(H)/\pi(H)$, so (63) follows from (62). □

The total area of cells meeting a bounded disk is finite:

$$\sum_{H \in \mathcal{H}(B_R(0))} a_H \leq \pi(R + D_R)^2. \tag{64}$$

Indeed the interiors are disjoint and every such cell is contained in B_{R+D_R}.

8.2 Area clock

Define

$$A(s) = \int_0^s \frac{a_{Y_u}}{m_0(Y_u)} \mathbf{1}_{\{Y_u \in V\}}\, du. \tag{65}$$

Theorem 8.2 (Area clock)**.** *Almost surely $A(s) < \infty$ for every finite s, A is continuous and strictly increasing, and $A(s) \to \infty$. Hence*

$$X_t = Y_{A^{-1}(t)} \tag{66}$$

is defined for all $t \geq 0$. At a vertex H, its holding time is exponential with mean $a_H/\pi(H)$ and the next vertex is K with probability $c(H,K)/\pi(H)$. The ordered reflected path and all graph-end labels are unchanged.

Proof. In one return cycle, the expectation of (65) restricted to cells meeting B_R is, by Lemma 8.1,

$$\frac{1}{\pi(o)} \sum_{H \in \mathcal{H}(B_R)} a_H < \infty.$$

Thus all integer-radius localized clocks are finite almost surely. Theorem 7.5 says that the vertex range of one finite return cycle is spatially bounded, so a random integer radius contains all cells visited in that cycle; its area-clock duration is therefore finite. Successive return times exhaust fast time, hence A is finite on compact intervals.

For each deterministic u, the fast process is vertex-valued almost surely; Fubini therefore implies that it is vertex-valued for Lebesgue-a.e. u along almost every sample path. Hence the clock density is positive a.e. and A is strictly increasing. Each holding interval at o is multiplied by $a_o/m_0(o)$; the resulting intervals are independent exponentials with positive mean $a_o/\pi(o)$, whose sum diverges. Thus $A(s) \to \infty$. The holding-time and transition assertions follow from deterministic rescaling of each vertex interval. □

The area-clock process is reversible with speed measure

$$a = \sum_H a_H \delta_H.$$

Indeed (65) is a positive continuous additive functional of the symmetric fast process with density $a_H/m_0(H)$ on vertices. The standard full-quasi-support time-change theorem therefore gives the same graph-energy form on $L^2(a)$,

$$\mathcal{F}_a = \{f \in L^2(a) : \mathcal{E}_G(f) < \infty\}, \qquad \mathcal{E}_a(f,g) = \mathcal{E}_G(f,g). \tag{67}$$

The later stochastic-calculus argument is carried out on a regular representation of the *fast* active reflected form supplied by the Silverstein-extension theory used by Gwynne–Sung [2, Proposition 3.15 and Section 3.6]; no measurable choice of such a representation across environments is needed.

8.3 Exact holding times

Let

$$h_H = \frac{a_H}{\pi(H)}.$$

On every complete holding interval of the fast ordered path, replace its duration by h_H, preserving the order and mapping linearly inside intervals.

Proposition 8.3 (Exact clock)**.** *The replacement defines a continuous increasing homeomorphism of the positive time axis. The resulting ordered path spends exactly h_H units of time at each visit to H, has spatially bounded vertex range on compact intervals, and avoids all spatial-infinity ends.*

Proof. In one return cycle, the expected exact duration contributed by cells meeting B_R is, by (63),

$$\sum_{H \in \mathcal{H}(B_R)} h_H \mathbb{E}_o N_H = \frac{1}{\pi(o)} \sum_{H \in \mathcal{H}(B_R)} a_H < \infty.$$

The random-radius argument used for the area clock gives local finiteness. The density on every holding interval is positive, those intervals cover almost every fast time, and repeated visits to o contribute the fixed positive duration h_o, so the clock is a homeomorphism. Spatial boundedness and end labels are invariant under such a time change. □

9 The reflected harmonic-coordinate martingale

9.1 Pure-jump structure of the reflected form

Fix the fast reflected form on $L^2(m_0)$ and a regular representation E in which the vertex set is a dense open discrete subset, as supplied by the standard regular representation of a Silverstein extension. All form functions below are identified with their quasi-continuous representatives on E.

Let

$$\eta_R(z) = \min\{1, (2 - |z|/R)_+\}.$$

It equals one on B_R, vanishes outside B_{2R}, and is R^{-1}-Lipschitz.

Lemma 9.1 (Bounded spatial localizers)**.** *For each $R > 0$ and $i = 1, 2$, the vertex functions*

$$\eta_R(z_H), \qquad \eta_R(z_H) z_H^i, \qquad \eta_R(z_H)\Phi^i(H) \tag{68}$$

are bounded and have finite graph energy. Hence they belong to the fast form domain and, where relevant, to the area form domain.

Proof. An edge on which one of the functions in (68) is nonconstant has an endpoint with representative in B_{2R}. By (30), both endpoint cells then lie in one bounded spatial patch. On this patch the representative functions and Φ are bounded; their graph energies are finite by Corollary 5.4 and (11). The product estimate

$$|\eta(H)f(H) - \eta(K)f(K)| \le |\eta(H)||f(H) - f(K)| + |f(K)||\eta(H) - \eta(K)|$$

and $|z_H - z_K| \le d_H + d_K$ finish the energy bound. Boundedness gives $L^2(m_0)$ since $m_0(V) < \infty$, while (64) gives local $L^2(a)$. □

Proposition 9.2 (No hidden continuous energy)**.** *In the Beurling–Deny decomposition of the regular fast reflected form, the strongly local and killing parts vanish. The jump measure is*

$$J = \frac{1}{2} \sum_{\{H,K\} \in E(G)} c(H, K)\big(\delta_{(H,K)} + \delta_{(K,H)}\big). \tag{69}$$

Consequently, for every bounded u in the fast form domain, the martingale additive functional in its Fukushima decomposition has no continuous martingale part and predictable bracket

$$\int_0^t \frac{1}{m_0(Y_s)} \sum_{K \sim Y_s} c(Y_s, K)(u(K) - u(Y_s))^2 \mathbf{1}_{\{Y_s \in V\}}\, ds. \tag{70}$$

The corresponding polarized formula holds for cross brackets.

Proof. The discrete measure (69) is Radon off the diagonal. Indeed, on any compact subset of $E \times E$ separated from the diagonal, finitely many continuous bounded finite-energy core functions separate the two coordinates with a uniformly positive squared difference; the J-mass is therefore bounded by a finite linear combination of their graph energies. For every bounded form-domain function u,

$$\int_{E \times E \setminus \mathrm{diag}} (u(x) - u(y))^2 J(dx, dy) = \mathcal{E}_G(u),$$

which is the entire quadratic form by Proposition 1.6(iv). Uniqueness of the Beurling–Deny decomposition forces the strongly local and killing parts to vanish and identifies J.

The Lévy-system/jump-compensator formula at a vertex gives (70). Since J has no mass on pairs with a nonvertex endpoint, there are no boundary jumps. The vanishing strongly local part gives no continuous martingale component. Properly exceptional sets have zero m_0-mass and therefore contain no vertex because every vertex has positive m_0-mass. □

9.2 Spatial and harmonic-coordinate extensions

Proposition 9.3 (Pathwise coordinate extensions)**.** *The vertex functions z_{Y_t} and $\Phi(Y_t)$ admit unique càdlàg extensions along the reflected path, continuous at every nonvertex time. Their only jumps are ordinary graph-edge jumps. The same holds after the area time change. At a time whose graph-end label is ω, the spatial extension equals p_ω.*

Proof. Fix an integer time horizon T. Theorem 7.5 bounds the vertex positions on $[0, T+1]$ by a random radius. For any sufficiently large deterministic cutoff radius R, the localized functions in Lemma 9.1 agree with the uncut coordinates at every vertex visited in this interval. Their quasi-continuous representatives evaluated on the Hunt path are càdlàg. Two sufficiently large cutoffs agree at every rational time, because fixed deterministic times are vertex-valued almost surely; right continuity gives agreement everywhere on $[0, T]$. This defines consistent local extensions as T grows.

Proposition 9.2 excludes jumps involving nonvertex states, so these coordinate paths are continuous at nonvertex times. Right-continuous extensions are unique because vertex-valued rational times are dense. The area clock is a continuous time homeomorphism. At an end time, Proposition 3.6 and Theorem 7.5 force the surrounding cells to converge to the finite spatial image p_ω. □

9.3 Finite-boundary localization

Lemma 9.4 (Finite-target extensions are in the generator domain)**.** *Let $A \subset V$ be nonempty and finite, let $\varphi : A \to \mathbb{R}$, and let h_A be the unique global finite-energy minimizer with trace φ on A. Then h_A belongs to the generator domain of the fast reflected form and*

$$Lh_A(H) = \begin{cases} \dfrac{1}{m_0(H)} \displaystyle\sum_{K \sim H} c(H,K)(h_A(K) - h_A(H)), & H \in A, \\ 0, & H \notin A. \end{cases} \tag{71}$$

The generator is bounded and supported on A.

Proof. The maximum principle makes h_A bounded, hence $h_A \in L^2(m_0)$, and it has finite energy. For v in the form domain let $v_A = \sum_{H \in A} v(H)\mathbf{1}_{\{H\}}$. Local finiteness makes v_A finite energy. Then $v - v_A$ vanishes on A, so full variational orthogonality gives

$$\mathcal{E}_G(h_A, v) = \mathcal{E}_G(h_A, v_A) = -\sum_{H \in A} m_0(H) Lh_A(H) v(H),$$

with Lh_A as in (71). The right side is the $L^2(m_0)$ pairing with a bounded finite-support function, which is precisely the generator-domain criterion. □

Theorem 9.5 (Harmonic coordinates through reflection times)**.** *The càdlàg process $M_t^0 = \Phi(Y_t)$ is a locally square-integrable local martingale. It has no continuous martingale part and*

$$\langle (M^0)^i, (M^0)^j \rangle_t = \int_0^t \frac{1}{m_0(Y_s)} \sum_{K \sim Y_s} c(Y_s, K) \Delta\Phi^i \Delta\Phi^j \mathbf{1}_{\{Y_s \in V\}}\, ds. \tag{72}$$

Proof. Choose an increasing sequence of good rectangles Q_n such that $B_n(0) \subset Q_n^\circ$, the starting cell is in Q_1°, and the union of all cells meeting Q_n is contained in Q_{n+1}°. This is possible by (30) and density of good line offsets. Let $A_n = \mathcal{H}(\partial Q_n)$; Proposition 3.3 makes A_n finite and nonempty. Let

$$\tau_n = \inf\{t \geq 0 : Y_t \in A_n\}.$$

The finite-separator property of the reflected path keeps the vertex path, until τ_n, in the component of $G \setminus A_n$ containing the start; this component lies in $G_{Q_n} \setminus A_n$. The spatial boundedness theorem gives $\tau_n \uparrow \infty$ almost surely.

For one coordinate i, let h_n be the global full-energy extension of $\Phi^i|_{A_n}$. On G_{Q_n}, both h_n and Φ^i have the same boundary trace and satisfy the full variational condition, so uniqueness in Lemma 6.2 gives $h_n = \Phi^i$ there. By Lemma 9.4, the Fukushima zero-energy term of $h_n(Y_t)$ is

$$\int_0^t Lh_n(Y_s)\, ds,$$

with integrand supported on A_n. Hence it vanishes up to τ_n. Therefore

$$\Phi^i(Y_{t\wedge\tau_n}) - \Phi^i(Y_0)$$

is a bounded square-integrable martingale. These stopping times localize $\Phi^i(Y)$.

Before τ_n, every neighbor of a visited cell remains in G_{Q_n}, so all edge differences of h_n equal those of Φ^i. Proposition 9.2 therefore identifies the stopped predictable bracket with (72). Let $n \to \infty$ and polarize. □

Theorem 9.6 (Area-clock reflected martingale)**.** *For X in Theorem 8.2, $M_t = \Phi(X_t)$ is a locally square-integrable local martingale with bracket (14). Its optional quadratic covariation is*

$$[M^i, M^j]_t = \sum_{0<s\le t} \Delta M_s^i \Delta M_s^j, \tag{73}$$

and there is no additional bracket contribution at reflection times.

Proof. Time-change the localized martingales in Theorem 9.5 by the inverse of A. On a vertex interval, $dt = (a_H/m_0(H))ds$, which converts (72) into (14). Proposition 9.2 says the fast martingale has no continuous martingale part and its optional bracket is the sum over ordinary graph jumps. Composition with a continuous increasing homeomorphism preserves this fact; in particular, an accumulation of holding intervals at a reflection time creates no singular continuous bracket. □

9.4 Continuous interpolation

On every holding interval $[s, t)$ at H whose next vertex is K, linearly interpolate from z_H to z_K over $[s, t]$; use the limits from Proposition 9.3 at end-valued times. The resulting path $\widetilde{Z}$ is continuous. Indeed, near an end-valued time the path leaves every fixed finite vertex set, while Theorem 7.5 keeps the visited cells in a bounded region. Lemma 5.1 then forces their diameters, and hence neighboring representative differences, to tend to zero. The same argument shows that any two choices of representatives have the same end-time limit.

10 A scale-free temporal ergodic theorem

The law of the environment is not stationary at a fixed Euclidean scale, so ordinary time-stationarity of the annealed process is unavailable. The correct replacement is a scale-covariant mass-transport identity on time, followed by the class-kernel lemma from Section 4.

10.1 Temporal mass transport

For a fixed environment, let $\mathbb{P}^v_{\mathcal{H}}$ be the two-sided law of the area-clock process with $X_0 = v$, with independent forward and time-reversed backward halves. Let

$$\mathbb{Q}_{\mathcal{H}} = \sum_{v \in V} a_v \mathbb{P}^v_{\mathcal{H}}. \tag{74}$$

By reversibility of the area process, $\mathbb{Q}_{\mathcal{H}}$ is invariant under deterministic time shifts. It is sigma-finite on the cover $\{X_0 = v\}$, each of finite mass a_v.

The annealed rooted trajectory Ω is obtained by first sampling $\mathcal{H}$ and then the two-sided process from H_0. When the current vertex changes, the environment is re-rooted at that cell; spatial translations are understood through any fixed measurable cell-rooted encoding. Write $\theta_t\Omega$ for time shift by t at vertex-valued times. A spatial dilation C acts parabolically by

$$\mathcal{S}_C\Omega = (C\mathcal{H}, (CX_{t/C^2})_{t\in\mathbb{R}}). \tag{75}$$

A functional is *unmarked* if it does not depend on an auxiliary time grid.

Lemma 10.1 (Temporal mass transport)**.** *Suppose $V(\Omega, s, t) \geq 0$ is measurable, covariant under time shifts,*

$$V(\theta_r\Omega, s - r, t - r) = V(\Omega, s, t),$$

and has parabolic scaling degree -2,

$$V(\mathcal{S}_C\Omega, C^2 s, C^2 t) = C^{-2} V(\Omega, s, t).$$

Then

$$\mathbb{E}\int_{\mathbb{R}} V(\Omega, 0, t)\, dt = \mathbb{E}\int_{\mathbb{R}} V(\Omega, t, 0)\, dt. \tag{76}$$

The identity remains valid after adjoining an independent time dyadic grid, provided the transport is covariant in the marked variables.

Proof. For unique source and target cells define the spatial transport

$$T(\mathcal{H}, w, z) = \frac{1}{a_{H_z}} \mathbb{E}^{H_w}_{\mathcal{H}} \int_{\mathbb{R}} \mathbf{1}_{\{X_t = H_z\}} V(\Omega, 0, t)\, dt,$$

and set it to zero at ambiguous roots. Under spatial dilation, a_{H_z} and dt each scale by C^2, while V scales by C^{-2}, so T has degree -2.

Integrating in z cancels $a^{-1}_{H_z}$ and gives the left side of (76). For the incoming integral, fix $o = H_0$. Integration over source cells gives

$$\frac{1}{a_o} \sum_v a_v \mathbb{E}^v_{\mathcal{H}} \int_{\mathbb{R}} \mathbf{1}_{\{X_t = o\}} V(\Omega, 0, t)\, dt.$$

For each deterministic t, shift the sigma-finite measure (74) by t to obtain

$$\mathbb{E}^o_{\mathcal{H}} \int_{\mathbb{R}} V(\Omega, -t, 0)\, dt,$$

which is the right side after the change of variable $s = -t$. Apply spatial mass transport and Tonelli. Independent dyadic marks may be averaged out first. □

10.2 Time blocks and literal reverse conditional expectations

The holding intervals of the two-sided area path form a countable disjoint family $\mathcal{I}$ covering Lebesgue-a.e. time. All have positive finite length.

Lemma 10.2 (Holding intervals are submacroscopic)**.** *Almost surely, for every $K > 0$,*

$$\#\{I \in \mathcal{I} : \operatorname{dist}(0, I) \le K|I|\} < \infty. \tag{77}$$

Consequently

$$\sup\{|I| : I \cap [-T, T] \neq \varnothing\} = o(T), \tag{78}$$

and the supremum is finite for every finite T.

Proof. Apply Lemma 10.1 to

$$V(\Omega, s, t) = |I_s|^{-1} \mathbf{1}_{\{\operatorname{dist}(t, I_s) \le K|I_s|\}}.$$

Its outgoing integral is $1 + 2K$, while its incoming integral counts the intervals in (77). Hence the count has finite expectation. An interval meeting $[-T, T]$ with length at least εT belongs to the fixed finite collection in (77) with $K \ge 1/\varepsilon$, which gives (78); the same argument gives finiteness of the supremum at fixed T. □

Let $\mathcal{D}_t$ be an independent uniform dyadic system on $\mathbb{R}$. For $J \in \mathcal{D}_t$ let $J^{(k)}$ denote the kth ancestor and set

$$D(J) = \sup_{I \cap J \neq \varnothing} |I|, \quad \alpha(J) = \sup_{k \ge 0} \frac{D(J^{(k)})}{|J^{(k)}|}, \quad \beta(J) = \alpha(J)^{-1}, \quad \kappa(J) = \sum_{k \ge 0} 4^{-k} \beta(J^{(k)}). \tag{79}$$

Lemma 10.2 gives the same monotonicity relations as (32). For $m > 0$ the maximal dyadic interval through s satisfying

$$\kappa(J) \le m < \kappa(J^{(1)})$$

defines, for almost every s, a nested covariant partition $J_m(s)$ of time. Its root blocks are finite and exhaust $\mathbb{R}$ as $m \to \infty$.

Write $\widehat{\Omega} = (\Omega, \mathcal{D}_t)$ and let $\widehat{\mathbb{P}}_t$ be the product of the annealed path law and the independent time-grid law. Let $\mathcal{A}_{\mathrm{par}}$ be the sigma-field of parabolically scale-invariant events of this marked trajectory. Define the uniform rerooting kernel

$$K_m^t(\widehat{\Omega}, A) = \frac{1}{|J_m(0)|} \int_{J_m(0)} \mathbf{1}_A(\theta_s \Omega, \mathcal{D}_t - s)\, ds, \qquad A \in \mathcal{A}_{\mathrm{par}}. \tag{80}$$

Proposition 10.3 (Temporal block projection)**.** *The kernels K_m^t satisfy Lemma 4.1 on the parabolically scale-invariant sigma-field. Therefore, for every integrable unmarked parabolically scale-invariant F,*

$$A_m := \frac{1}{|J_m(0)|} \int_{J_m(0)} F(\theta_s \Omega)\, ds = \mathbb{E}[F \mid \mathcal{G}_m^t], \tag{81}$$

and the sigma-fields $\mathcal{G}_m^t$ decrease with m.

Proof. Rerooting inside a block gives the same translated block and hence the same uniform class kernel. For stationarity, let $U \ge 0$ be parabolically scale invariant and use Lemma 10.1 with

$$V(\Omega, \mathcal{D}_t, s, t) = |J_m(s)|^{-1} \mathbf{1}_{\{t \in J_m(s)\}} U(\theta_t \Omega, \mathcal{D}_t - t).$$

The incoming integral is U and the outgoing integral is $K_m^t U$. Hence $\int K_m^t U \, d\widehat{\mathbb{P}}_t = \int U \, d\widehat{\mathbb{P}}_t$ on $\mathcal{A}_{\mathrm{par}}$. Lemma 4.1 gives (81). Nesting of the time partitions gives the decreasing sigma-fields. □

Lemma 10.4 (Complete dyadic-chain convergence)**.** *For integrable F as above, A_m converges almost surely and in L^1, through positive rational m, to*

$$L = \mathbb{E}\left[F \mid \bigcap_{n\geq 1} \mathcal{G}_n^t\right]. \tag{82}$$

The same limit holds along the complete root dyadic ancestor chain. The process $t \mapsto F(\theta_t\Omega)$ is locally integrable almost surely. The limit has a version invariant under every vertex-valued time rerooting and under common parabolic scaling of trajectory and grid.

Proof. The integer-parameter reverse martingale theorem gives almost-sure and L^1 convergence. If rational $m \geq n$, the tower property gives

$$A_m - L = \mathbb{E}[A_n - L \mid \mathcal{G}_m^t].$$

Doob's weak-L^1 inequality applied to finite rational lists, followed by exhaustion, yields

$$\mathbb{P}\left[\sup_{m\in\mathbb{Q},\, m\geq n} |A_m - L| > \delta\right] \leq \frac{\mathbb{E}|A_n - L|}{\delta} \longrightarrow 0.$$

The events decrease in n, giving convergence over all rational parameters. Every root dyadic interval is selected on a nonempty interval of parameters containing a rational, so the complete ancestor chain has the same limit. Apply the argument to $|F|$ to get local integrability. Finally, sufficiently large root blocks contain any fixed pair of vertex times in the same interior block; their averages are unchanged by rerooting. Scaling also preserves the averages. This gives the invariant version. □

10.3 Regeneration removes the remaining invariant randomness

Lemma 10.5 (Regenerative invariant functionals)**.** *Fix one environment and its recurrent reversible area-clock process. Let B be a bounded measurable unmarked functional of the two-sided path which is invariant under every vertex-valued time rerooting. Then B is almost surely constant under $\mathbb{P}^v_{\mathcal{H}}$ for every v, and the constant is independent of v. If B also transforms covariantly under spatial translations and parabolic scaling, this common constant is a measurable similarity-invariant function of $\mathcal{H}$.*

Proof. Fix v and concatenate a doubly infinite i.i.d. sequence of complete return cycles from v, with time zero at the entrance of the zeroth cycle. The cycle-shift is a Bernoulli shift. Since B is invariant under rerooting from one cycle entrance to the next, it is invariant under this shift and therefore constant almost surely.

To compare this entrance-rooted law with $\mathbb{P}^v_{\mathcal{H}}$, bias the zeroth cycle by the length of its initial holding interval and choose time zero uniformly in that interval. The initial holding time is exponential with mean $h_v = a_v/\pi(v)$ and is independent of the excursion. Under this length bias and uniform choice, the age and residual lifetime are independent exponentials of mean h_v. Reversibility of the killed semigroup at v implies that the excursion law from departure until the next hit of v is invariant under time reversal. Thus the past and future obtained by this Palm shift are exactly the independent time-reversed and forward halves of $\mathbb{P}^v_{\mathcal{H}}$. The biased law is absolutely continuous with respect to the entrance-rooted cycle law, and moving the root inside the holding interval does not change B, so B is constant under $\mathbb{P}^v_{\mathcal{H}}$.

If u, v are vertices, the shift-invariant sigma-finite measure (74) and irreducibility show that the two constants agree: for some deterministic $t > 0$, the event $\{X_t = u\}$ has positive probability

under $\mathbb{P}^v_{\mathcal{H}}$ by following a fixed finite graph path quickly and then holding at u. On that event time-shift invariance identifies the constants. Measurability follows from the measurable conditional path law. Similarity covariance of B and of the area-clock kernel makes the common constant a similarity-invariant environment function. □

Proposition 10.6 (Scale-free temporal ergodic theorem)**.** *If $F(\Omega)$ is integrable, unmarked, and parabolically scale invariant, then almost surely*

$$\lim_{T\to\infty} \frac{1}{T}\int_0^T F(\theta_t\Omega)\,dt = \mathbb{E}[F]. \tag{83}$$

The corresponding backward average has the same limit.

Proof. Let L be the dyadic-chain limit in Lemma 10.4. It may still depend on the independent time grid. For bounded Borel ψ, define

$$B_\psi(\Omega) = \mathbb{E}_{\mathcal{D}_t}[\psi(L(\Omega, \mathcal{D}_t))].$$

The invariant version of L and invariance of the grid law make B_ψ invariant under all vertex rerootings and parabolic scaling. Lemma 10.5 makes it a similarity-invariant function of the environment; environment ergodicity makes it deterministic. Its value is necessarily $\mathbb{E}[\psi(L)]$.

Since F is unmarked and the grid is independent,

$$\mathbb{E}[F\psi(L)] = \mathbb{E}[F]\mathbb{E}[\psi(L)].$$

Since L is the reverse conditional-expectation limit of F,

$$\mathbb{E}[F\psi(L)] = \mathbb{E}[L\psi(L)].$$

Hence $\mathbb{E}[(L - \mathbb{E}F)\psi(L)] = 0$ for every bounded ψ, and a bounded sign test gives $L = \mathbb{E}F$ almost surely.

It remains to pass from dyadic ancestor intervals to $[0, T]$. First take $F \geq 0$ and set $\mu = \mathbb{E}F$. If $T^{-1}\int_0^T F(\theta_t\Omega)dt > \mu + \varepsilon$ for arbitrarily large T, local integrability permits a measurable rational witness $T_n \geq n$ whenever such a witness exists. Choose $\delta \in (0, 1)$ with $(\mu+\varepsilon)/(1+\delta) > \mu+\varepsilon/2$. For an independent uniform dyadic grid, conditional on the path, the probability that an origin dyadic interval contains $[0, T_n]$ and has length in $[T_n, (1 + \delta)T_n]$ is a strictly positive constant p_δ depending only on δ. On this event the dyadic average exceeds $\mu + \varepsilon/2$. Complete-chain convergence therefore forces the probability of a forward witness beyond n to tend to zero. Thus the forward limsup is at most μ.

For bounded $F \geq 0$, apply the same upper bound to $K - F$ to obtain the lower bound. For general nonnegative integrable F, truncate by $F \wedge K$ and let $K \to \infty$; then split a real integrable F into positive and negative parts. The backward-time proof is identical. □

11 Bracket asymptotics and the invariance principle

Put Γ as in (14) and Σ as in (15). Every entry of Γ is an unmarked parabolically scale-invariant functional of the rooted area-clock environment: under a spatial dilation by C, both a_H and $|\Delta\Phi|^2$ are multiplied by C^2. Its entries are integrable by (12) and $|xy| \leq (x^2 + y^2)/2$.

11.1 Deterministic bracket and nondegeneracy

Lemma 11.1 (Deterministic bracket limit)**.** *The matrix Σ is finite, symmetric, and positive definite. Conditionally on almost every environment, for the area-clock martingale started from H_0,*

$$\frac{1}{T}\langle M^i, M^j\rangle_T \longrightarrow \Sigma_{ij} \qquad \text{almost surely as } T \to \infty. \tag{84}$$

Consequently, if $M^\varepsilon_t = \varepsilon M_{t/\varepsilon^2}$, then

$$\langle (M^\varepsilon)^i, (M^\varepsilon)^j\rangle_t \longrightarrow t\Sigma_{ij} \tag{85}$$

in quenched probability, uniformly for t in compact intervals.

Proof. Apply Proposition 10.6 to each entry of Γ and use Theorem 9.6. This gives the annealed almost-sure limit; Fubini gives (84) for almost every fixed environment. For diagonal entries, uniformity on compact time intervals follows from monotonicity and a finite mesh; polarization gives the cross terms.

It remains to prove positive definiteness. Suppose that a deterministic $e \in \mathbb{R}^2 \setminus \{0\}$ satisfies $e^{\mathsf{T}}\Sigma e = 0$. Then the nonnegative rooted specific-energy density of $e \cdot \Phi$ has expectation zero. Proposition 5.3, applied to that density, implies that the gradient of $e \cdot \Phi$ vanishes on every edge almost surely. Global graph connectedness therefore gives $e \cdot \Phi \equiv 0$ after the normalization at H_0.

Fix a realization in this full-probability event. For large R, choose

$$x_R \in B_1(Re/|e|) \setminus S$$

for an admissible witness S, and let H_R be a cell containing x_R. Since H_R meets $B_{R+1}(0)$,

$$|z_{H_R} - Re/|e|| \le 1 + D_{R+1} = o(R)$$

for any representative $z_{H_R} \in H_R$, by Lemma 5.1. By (13), $|\Phi(H_R) - z_{H_R}| = o(R)$. Hence

$$e \cdot \Phi(H_R) = R|e| + o(R),$$

contradicting $e \cdot \Phi \equiv 0$. Thus Σ has no nonzero deterministic null vector. Since it is deterministic, symmetric, and nonnegative definite, it is positive definite. □

11.2 Compact containment and large jumps

Lemma 11.2 (Coercivity and uniformly small possible jumps)**.** *Almost surely there is $K_0 < \infty$ such that*

$$|z_H| \le 2|\Phi(H)| + K_0 \qquad (H \in \mathscr{H}). \tag{86}$$

Moreover, for every fixed $R < \infty$,

$$\lim_{\varepsilon \downarrow 0} \sup_{|z_H| \le R/\varepsilon} \sup_{K \sim H} \varepsilon|\Phi(K) - \Phi(H)| = 0. \tag{87}$$

Proof. For a vertex with $r = |z_H|$ large, H meets $B_r(0)$, so uniform sublinearity gives $|\Phi(H) - z_H| \le r/2$. Enlarging the constant over the cells in a fixed bounded region gives (86).

For (87), suppose $|z_H| \le L$. Then H lies in $B_{L+D_L}(0)$, and every neighbor K meets that disk because $H \cap K \ne \varnothing$. Since $D_R/R \to 0$, for large L both z_H and every such z_K lie in $B_{2L}(0)$. Uniform sublinearity gives

$$\sup_{|z_H| \le L,\ K \sim H} |(\Phi(K) - z_K) - (\Phi(H) - z_H)| = o(L),$$

and (4), or the representative-independent form of Lemma 5.1, gives $|z_K - z_H| = o(L)$ uniformly in the same range. Take $L = R/\varepsilon$. □

Lemma 11.3 (Quenched compact containment and Lindeberg condition)**.** *For every $T < \infty$, conditionally on almost every environment, the families*

$$\left\{ \sup_{0\le t\le T} |M_t^\varepsilon| : \varepsilon \in (0,1) \right\}, \qquad \left\{ \sup_{0\le t\le T} \varepsilon |Z_{t/\varepsilon^2}| : \varepsilon \in (0,1) \right\} \tag{88}$$

are tight. For every $\delta > 0$,

$$\varepsilon^2 \int_0^{T/\varepsilon^2} \frac{1}{a_{X_s}} \sum_{K\sim X_s} c(X_s,K) |\Phi(K) - \Phi(X_s)|^2 \mathbf{1}_{\{\varepsilon|\Phi(K)-\Phi(X_s)|>\delta\}}\, ds \longrightarrow 0 \tag{89}$$

in quenched probability.

Proof. For the vector local martingale M^ε, $|M^\varepsilon|^2 - \mathrm{tr}\langle M^\varepsilon\rangle$ is a local martingale. Stop when the increasing compensator $\mathrm{tr}\langle M^\varepsilon\rangle$ first reaches L and use the square-integrable submartingale maximal inequality. For a numerical constant which we absorb into L,

$$\mathbb{P}\left[\sup_{t\le T} |M_t^\varepsilon| > K \mid \mathcal{H} \right] \le \frac{L}{K^2} + \mathbb{P}\left[\mathrm{tr}\langle M^\varepsilon\rangle_T \ge L \mid \mathcal{H} \right]. \tag{90}$$

Lemma 11.1 makes the second term small uniformly for small ε once L is large, and then the first term is small for large K. This proves compact containment for M^ε. Equation (86), first at vertex times and then by the extension of Proposition 9.3, transfers it to $\varepsilon Z_{t/\varepsilon^2}$.

Fix R. On the event

$$\sup_{0\le t\le T} \varepsilon |Z_{t/\varepsilon^2}| \le R,$$

all possible graph jumps from states visited before T/ε^2 have scaled harmonic-coordinate size $o(1)$ uniformly, by (87). Hence the integrand in (89) is identically zero for all sufficiently small ε. Compact containment lets $R \to \infty$ after $\varepsilon \to 0$. □

11.3 Martingale FCLT and spatial path

We use the continuous-predictable-bracket case of the martingale functional central limit theorem in Whitt [5, Theorem 2.1(ii)]. In the form needed here, convergence of predictable brackets to $t\Sigma$, together with vanishing expected squared maximal martingale jump and vanishing maximal jump of the predictable bracket, implies convergence in the Skorokhod J_1 topology to Brownian motion with covariance Σ.

Theorem 11.4 (Area-clock Brownian limit)**.** *Conditionally on almost every environment,*

$$(M_t^\varepsilon)_{t\ge 0} \Longrightarrow (B_t)_{t\ge 0} \tag{91}$$

in the Skorokhod J_1 topology, where B has covariance Σ. The continuously interpolated spatial paths satisfy (16)*.*

Proof. Fix a good environment. By (87), choose deterministic numbers $\varepsilon_n \downarrow 0$ such that whenever $0 < \varepsilon \le \varepsilon_n$, every possible scaled harmonic-coordinate jump from a vertex satisfying $|z_H| \le n/\varepsilon$ has size at most $1/n$. Choose the sequence strictly decreasing and set

$$r_\varepsilon = \max\{n : \varepsilon \le \varepsilon_n\}.$$

Then $r_\varepsilon \to \infty$. Stop M^ε at

$$\tau_\varepsilon = \inf\{t \geq 0 : \varepsilon |Z_{t/\varepsilon^2}| \geq r_\varepsilon\}.$$

Lemma 11.3 gives

$$\mathbb{P}[\tau_\varepsilon \leq T \mid \mathcal{H}] \longrightarrow 0$$

for every fixed T. Every graph jump of the stopped martingale, including the possible exit jump, has size at most $1/r_\varepsilon$; a continuous exit adds no jump. Its predictable bracket is continuous and agrees with that of the unstopped martingale on $[0, T]$ whenever $\tau_\varepsilon > T$. Lemma 11.1 therefore gives bracket convergence, while

$$\mathbb{E}\left[\sup_{0<s\leq T} |\Delta M^\varepsilon_{s\wedge\tau_\varepsilon}|^2 \mid \mathcal{H}\right] \leq r_\varepsilon^{-2} \longrightarrow 0.$$

Whitt's theorem gives Brownian convergence for the stopped martingales. Since the probability of stopping tends to zero on every compact interval, (91) follows.

By compact containment, with quenched probability tending to one the spatial path through time T/ε^2 stays in a ball $B_{R/\varepsilon}(0)$, with R fixed. Uniform sublinearity then gives

$$\sup_{0\leq t\leq T} \varepsilon |Z_{t/\varepsilon^2} - M_{t/\varepsilon^2}| \longrightarrow 0 \tag{92}$$

in quenched probability. Hence the càdlàg spatial path has the same J_1 limit.

Finally, an interpolation segment across $H \sim K$ differs from the step path by at most $|z_H - z_K| \leq d_H + d_K$. On a spatially contained path, Lemma 5.1 makes the largest such scaled error $o(1)$. Thus the continuous interpolation has the same limit. Since the limit is continuous, J_1 convergence of the interpolated paths is equivalent to local uniform convergence. □

11.4 Deterministic holding times

Let

$$h_H = \frac{a_H}{\pi(H)}. \tag{93}$$

Consider the stationary two-sided exponential area-clock path. If I_s is the holding interval containing s, let $L(I_s)$ be its total length. At non-holding times put the expressions below equal to zero. Define

$$V(\Omega) = \frac{h_{X_0}}{L(I_0)}, \qquad C(t) = \int_0^t \frac{h_{X_s}}{L(I_s)}\, ds. \tag{94}$$

Lemma 11.5 (Asymptotic identity of the exact clock)**.** *Almost surely,*

$$\frac{C(t)}{t} \longrightarrow 1 \qquad (t \to \infty). \tag{95}$$

The function C is a continuous strictly increasing homeomorphism of $[0, \infty)$ onto itself and, for each $T < \infty$,

$$\sup_{0\leq t\leq T} \left|\varepsilon^2 C^{-1}(t/\varepsilon^2) - t\right| \longrightarrow 0. \tag{96}$$

Proof. The functional V is unmarked and parabolically scale invariant. Conditional on the current cell H, the age and residual life of the stationary exponential holding interval are independent exponentials of mean h_H. Hence $L(I_0)$ has density

$$\ell h_H^{-2} e^{-\ell/h_H}\, d\ell,$$

and therefore

$$\mathbb{E}\left[\frac{h_H}{L(I_0)}\,\middle|\,H\right] = 1.$$

Proposition 10.6 gives (95). On each complete holding interval I at H, the integral defining C increases by exactly h_H. Its density is positive almost everywhere, local finiteness follows from Proposition 8.3, and repeated returns to a fixed vertex force divergence. Thus C is a homeomorphism. The inverse estimate (96) follows from monotonicity and (95). □

Theorem 11.6 (Exact holding times)**.** *The reflected ordered path in which every visit to H has deterministic duration $h_H = a_H/\pi(H)$ has the same Brownian limit (16), with the same matrix Σ.*

Proof. Apart from the initial holding interval, the exact-clock path is the exponential area-clock path composed with C^{-1}. By (96), after diffusive rescaling this random time change converges locally uniformly to the identity. Theorem 11.4 and the continuous mapping theorem therefore give the same Brownian limit. If at time zero the exact-clock construction assigns a full deterministic holding interval instead of only the residual portion inherited from the stationary two-sided path, the discrepancy is at most the fixed number h_{H_0} and vanishes after multiplication of time by ε^2. Linear interpolation is preserved by the monotone reparametrization. □

11.5 Starting vertex and representative

Proposition 11.7 (Fixed starting cells and arbitrary representatives)**.** *The conclusions of Theorems 11.4 and 11.6 hold from every fixed starting cell and for every measurable choice of a representative point in each cell.*

Proof. Fix a good environment and a vertex v. Recurrence gives an almost surely finite first hit of H_0. The spatial range and the elapsed area or exact time before this hit are almost surely finite. Under diffusive rescaling both the initial duration and the initial spatial displacement vanish in probability. After the hit, the area-clock process has the law from H_0 by the strong Markov property; for the exact clock the same ordered-path decomposition holds at vertex arrivals. This transfers the limit to v. Since every environment has only countably many cells, one full-probability environment set works for every fixed cell.

If $z_H, z'_H \in H$, then $|z_H - z'_H| \le d_H$. On a path contained in $B_{R/\varepsilon}(0)$, Lemma 5.1 gives $\varepsilon \sup d_H \to 0$. Compact containment therefore makes the two scaled interpolated paths asymptotically identical. At reflection times Proposition 9.3 gives the same limiting spatial value. □

Theorems 8.2, 9.6, 11.4, 11.6, and Proposition 11.7 prove Theorem 1.5.

12 Specialization to Gwynne–Miller–Sheffield

This section verifies, rather than merely asserts, the two specializations advertised in the introduction.

12.1 The nonsingular cell theorem

Corollary 12.1 (GMS Theorem 1.16)**.** *Suppose the singular witness can be taken to be empty. Then the hypotheses of Theorems 1.4 and 1.5 contain the hypotheses of GMS Theorem 1.16, with the same cell topology, the same moment condition, the same exact holding times, and the same quenched local-uniform Brownian conclusion. In this case the ordinary conductance walk is recurrent, so the reflected ordered path is the ordinary walk.*

Proof. If $S = \varnothing$, Definition 1.1 says that the cells cover $\mathbb{C}$, are spatially locally finite, compact and connected with nonempty interiors, and have pairwise zero-area intersections. The adjacency relation is any symmetric relation between distinct intersecting cells, with positive symmetric conductances. These are precisely the geometric objects in GMS Definition 1.15. Condition (LCS) is precisely their connectedness-along-horizontal-and-vertical-segments hypothesis. Proposition 2.5 identifies the environment topology and its Borel structure with theirs, and Corollary 2.9 identifies mass transport and ergodicity on this subspace. The moment (FE) is literally their two diameter–conductance moments combined into one expectation.

It remains only to identify the reflected ordered path with the ordinary recurrent walk. Let f_n be the logarithmic cutoff of (56), and put $u_n = 1 - f_n$. Then $u_n(o) = 1$, $\mathcal{E}(u_n) = \mathcal{E}(f_n) \to 0$, and u_n has finite vertex support because the configuration is spatially locally finite and u_n vanishes outside a bounded Euclidean disk.

Let (W_j) be the ordinary discrete-time conductance walk from o. For a finite set $K \ni o$, put

$$\sigma_o^+ = \inf\{j \geq 1 : W_j = o\}, \qquad \sigma_{K^c} = \inf\{j \geq 0 : W_j \notin K\}.$$

Let v_K minimize graph energy among functions equal to 1 at o and 0 on $V \setminus K$. The finite Dirichlet principle and the Markov property identify v_K with the probability of hitting o before exiting K. Testing the minimizer against $v_K - \mathbf{1}_{\{o\}}$ gives the standard capacity identity

$$\pi(o)\mathbb{P}_o(\sigma_{K^c} < \sigma_o^+) = \mathcal{E}(v_K). \tag{97}$$

Choose K containing $\operatorname{supp} u_n$. Minimality gives $\mathcal{E}(v_K) \leq \mathcal{E}(u_n)$, hence the left side of (97) tends to zero. On the event of no return to o, the walk exits every finite K almost surely: while it remains in a fixed finite set, positivity of the transition probabilities along finitely many paths gives a uniform positive probability, in a bounded number of steps, either to hit o or leave the set. Iteration rules out remaining forever without either event. Therefore $\mathbb{P}_o(\sigma_o^+ = \infty) = 0$. Irreducibility gives recurrence.

The recurrent ordinary process is the recurrent case of Proposition 1.6; no reflection at graph infinity occurs. The exact holding time in Theorem 11.6 is $a_H/\pi(H)$, exactly the GMS parametrization. Thus Theorems 1.4 and 1.5 give GMS Theorem 1.16. □

12.2 Embedded lattices and faces

Corollary 12.2 (GMS Theorem 1.5)**.** *Let M be a random embedded lattice with conductances satisfying the two hypotheses preceding GMS Theorem 1.5, with its law on the GMS embedded-lattice topology. Let X be the conductance walk on faces, started from the origin-containing face, and give every visit to a face H the deterministic duration* $\operatorname{Area}(H)/\pi(H)$*. For arbitrary measurable representatives $z_H \in H$,*

$$(\varepsilon z_{X_{t/\varepsilon^2}})_{t\geq 0} \Longrightarrow (B_t)_{t\geq 0}$$

quenched in the local-uniform topology, for a deterministic positive-definite covariance matrix.

Proof. Take the cells to be the closures of the faces of M. Two distinct faces are adjacent precisely when they share a primal edge, and put

$$c(H,K) = \sum_{e \text{ shared by } H,K} c_M(e). \tag{98}$$

This is exactly the face conductance used by GMS. Thus $\pi(H)$ is the total aggregated conductance out of the face, while

$$\pi^*(H) = \sum_{K\sim H} c(H,K)^{-1}$$

takes the reciprocal only after all primal-edge conductances shared by the pair (H, K) have been aggregated.

No new comparison between the embedded-lattice topology and the cell topology is needed here. GMS prove in Section 3.4 of [1] that their embedded-lattice Theorem 1.5 is a specialization of their cell theorem, Theorem 1.16: the face closures form a spatially locally finite full-plane cell configuration, the face-adjacency graph satisfies the horizontal/vertical segment-connectivity hypothesis, and the induced law satisfies the corresponding invariance and moment assumptions. Corollary 12.1 has already proved that, on precisely this nonsingular cell class, our topology, Borel structure, mass-transport/ergodicity hypotheses, moment condition, and exact clock agree with those of GMS Theorem 1.16. Therefore the specialization from GMS Section 3.4 may be composed directly with Corollary 12.1.

The resulting exact holding time is $\mathrm{Area}(H)/\pi(H)$, and Proposition 11.7 allows arbitrary measurable representatives in the faces. This is the conclusion of GMS Theorem 1.5. □

Remark 12.3 (What the singular theorem changes)**.** *Relative to the nonsingular GMS theorem, the proof changes only where spatial accumulation can occur. The replacements are: the cutoff whole-cell topology of Section 2; diameter-indexed spatial blocks and the projection lemma of Sections 4–5; full finite-energy minimization on possibly infinite spatial restrictions; the logarithmic cutoff argument excluding spatial infinity; and the reflected-path/area-clock analysis. The adjacency rule itself is not strengthened. In particular, intersecting cells need not be adjacent and no auxiliary exceptional set of primal vertices is introduced.*

13 Measurability and scope

We collect the measurable-structure points that are actually needed. This section is intentionally shorter than in the previous draft because the projection lemma removes the need to construct or analyze ad hoc invariant sigma-fields.

Proposition 13.1 (Measurability of the constructions)**.** *All environment quantities and kernels used above admit versions measurable for $\mathcal{B}_{\mathrm{sing}}$, without making a measurable choice of admissible witness and without choosing the Dirichlet-form regular representation measurably across environments.*

Proof. Use the auxiliary countable cell coding of Proposition 2.8. Area, centroid, and diameter are Borel functions of a compact set; adjacency and conductance coordinates are Borel by construction. Intersections with rectangles, line segments, and rational balls are Borel tests. Countable sums over cells or edges therefore give π, π^*, energies, the diameter bounds, and the line-variation quantities measurably.

The dyadic systems have their usual countable-coordinate standard Borel spaces. Selected blocks are defined by countably many inequalities involving their ancestors, so the block partitions and rerooting kernels are measurable. The class-kernel lemma requires only those kernels and the corresponding countably generated invariant sub-σ-algebras; no measurable equivalence-class selector is used.

For a countable locally finite weighted graph, finite Dirichlet problems are measurable finite linear systems. The full-energy minimizer of Lemma 6.2 is obtained as the pointwise limit of a measurable increasing sequence of boundary-anchored finite problems. The corrector limit is taken along deterministic subsequences and is characterized uniquely by its unmarked variational property, so Proposition 6.8 gives an unmarked measurable gradient and potential.

After graph local finiteness is known, the reflected construction uses countable network data, finite-target harmonic measures, countable transition choices, exponential variables, and countable

limits. This yields a measurable conditional law of the ordered vertex path. The spatial and harmonic-coordinate extensions are determined by their rational-time vertex values and pathwise right continuity. The auxiliary regular representation used in Section 9 is needed only to prove properties of a fixed quenched path law; it need not be selected measurably as the environment varies.

Finally, the witness S appears only in deterministic geometric implications after an environment has been fixed in Ω_{geom}. Every random quantity retained in the theorem—including end images, after Proposition 3.6, and the corrector—is independent of the choice of witness. Thus Assumption 1.2 is sufficient and no existential-witness Borel theorem is being invoked implicitly. □

Remark 13.2 (No joint-grid ergodicity)**.** *The proof never claims that an independent dyadic grid adjoined to an ergodic scale-free environment is jointly ergodic. Such a claim is false in general because the relative logarithmic phase of the grid and environment can be invariant. The only facts used are stationarity of the block rerooting kernels on the appropriate scale-invariant sigma-algebras, the class-kernel conditional-expectation lemma, and environment ergodicity after the auxiliary grid has been averaged out.*

14 Logical dependency summary

For clarity, the proof can be read in the following order.

(1) Section 2 fixes the environment topology and its exact GMS specialization.

(2) The geometric lemmas give global graph connectedness, graph local finiteness, submacroscopic cells, and spatial images of ends.

(3) Lemma 4.1 is applied twice: spatially to control specific-energy fields, and temporally to obtain a literal reverse martingale. This is the only abstract averaging mechanism in the paper.

(4) Sections 6 and 7 construct the harmonic coordinate, prove uniform sublinearity, and exclude spatial infinity.

(5) Sections 8 and 9 identify the area clock and prove that the harmonic coordinate remains a pure-jump local martingale through reflection times.

(6) Section 10 turns bracket densities into deterministic time averages. Section 11 applies the martingale FCLT and treats deterministic holding times.

(7) Section 12 checks the two GMS specializations assumption by assumption.

No finite-volume approximation of the environment law is asserted for genuinely accumulating configurations, no measurable singular-set mark is introduced, and no topology on labeled rational cells replaces the geometric whole-cell topology.

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